Fundamentals of IoT

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Fundamentals of IoT

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First Edition - 2016

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Preface

The book 'Fundamentals of IoT' serves to impart an understanding on the network protocols and connected devices in IoT. It will also explain the role of IoT in real-world applications such as the wearables, smart homes, and even smart cities. Additionally it will also explain security and privacy concerns with regards to IoT.

This book is the result of a concentrated effort of the Design Team, which is continuously striving to bring you the best and the latest in Information Technology. The process of design has been a part of the ISO 9001 Certification for Aptech-IT Division, Education Support Services. As part of Aptech's quality drive, this team does intensive research and curriculum enrichment to keep it in line with industry trends.

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Table of Content

Session Name

1. Introduction to Internet of Things
2. IoT in the Real World
3. Design and Strategy for Connected Devices
4. Security and Privacy Concerns in IoT

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SESSION 1- INTRODUCTION TO INTERNET OF THINGS

Session Objectives:

This session explains the concept of Internet of Things (IoT) and describes in detail its history, the objects used in IoT, and the role of Cloud Computing and Big-data in IoT.

At the end of this session, students will be able to:

- Explain IoT
- Enumerate the history of IoT
- List the objects in IoT
- Describe the role of Cloud Computing in IoT
- Describe the role of Big-data in IoT

1.1 Introduction to IoT

Internet of Things (IoT) is a rapidly evolving concept and is the base for future technology and innovations. The potential of IoT is endless and its need in today's world is inevitable. IoT is a virtual world, wherein, all living and non-living things are assigned with Unique Identifiers (IDs) and the ability to automatically transmit data to any network with or without human assistance. IoT makes daily life objects smarter; for instance, a SMART clip capable of giving alerts on the possibility of rain, with an automated message to the user's smartphone; so that, the clothes can be dried indoor as per the predicted rain.

IoT is a path to the smart-world with pervasive computing and networking. Its objective is to make varied tasks simpler for users and offer supplementary tasks, such as easy controlling and observing of various occurrences surrounding the user. Universal computing facilitates IoT to be embedded universally and automated, to act spontaneously without manual triggering. Moreover, pervasive computing is universal and provides technology and services across the globe - anywhere, any place and at any time.

Figure 1.1 shows the world of IoT



Figure 1.1: Depicts the World of Internet of Things

Environmental and everyday life items are characterized as 'objects', 'machines', or 'things'. To connect to the communication framework, these life items are enhanced with computing and advanced communication technology in the IoT. In this context, wired and wireless technologies already provide the communication abilities and connections, offering wide range of services based on individual-to-individual, individual-to-machine, machine-to-individual, machine-to-machine interactions, and so on. These types of connected machines, objects, or things act as an innovative Internet facility for network users and help in building data traffic in the emergent or growing Internet world. For example, a SMART car can sense the low fuel level and indicate or alarm the user for refuelling.

1.1.1 Smart Technology

The purpose of the IoT is to make objects smarter. In our regular life, most of the devices are electrified. Presently, the world around us is witnessing rapid electrification and immense gains from it. Electrification is defined as the process of powering by electricity. Products that are manufactured in the present world, are manufactured with an embedded processor, (usually a microcontroller), along with user interfaces which can complement a logical control functionality as well as programmability. The electrified world and the extensive usage of embedded processing make objects smarter.

The Microcontroller Unit (MCU) has turned the conventional toaster smart, which is capable of controlling the color of the toast automatically. It completes the task steadily and consistently, since, it is a smart toaster and has the ability to connect automatically through touchpad or switches. Once a device turns smart with the implanting of an embedded processor, logically, the next step is to remotely access these devices from a distance location, which will make life much easier and comfortable for users. For instance, the household devices such as lights can be controlled from the work place using smartphone.

1.1.2 Use of IoT in Real Life Technology

IoT is the future of real life technology. IoT has innumerable scope in real life uses. IoT has a significant place in the listed ranges to improve users' life as follows:

- Machine-to-Infrastructure communication
- System-to-System communication
- Tele-health: remote or real-time persistent observing of patients, diagnosis, and drug delivery
- Automated traffic management
- Home and industrial building automation
- Continuous checking and firmware improvements of vehicles
- Remote security and control
- Asset tracking of goods on the move
- Smart applications of IoT includes agriculture, grid, water, meters, buildings, broadband, cities, cars, appliances, tags, animal farming, and the environment
- Environmental monitoring and control

Figure 1.2 shows few examples of IoT in real life technology.



Figure 1.2: Examples of IoT in Real Life Technology

1.1.3 IoT and Quality of Life of Users

Quality of users' life has improved in numerous ways with smart IoT devices. Banking or shopping has become easy with a single click. IoT will improve the standards of users' life. For example, if a stranger enters into the house in the absence of owner of the house, the intelligent home intruder sensor identifies the presence of an unauthorized person. It alerts the security guards automatically and nearest on-duty police personnel. The home intruder sensor also triggers the surveillance and security cameras around the house and residential area, which is attached with advanced face and plate number identification technology that is linked to the national criminal database. It also communicates to the owner, a comprehensive report that comprises the visual along with the details of incident, on time. A custom-made report is sent to the local police station and the insurance company who is responsible for the burglary insurance (loss or damage to the insured property due to burglary or house-breaking).

This advanced smart imaginary home situational response system helps the house owner to relax and be stress-free; reduces the paperwork and upgrades the quality of life.

1.1.4 Moving towards Ubiquitous or Pervasive Computing

Researchers developed a human-to-human interface over technology in late 1980s, which resulted in the design of the universal computing technology that focuses to embed technology into users' daily life. After the Personal Computer (PC) age, new handheld devices and smartphones are changing users' environment by turning it more communicative and instructive. The 'forefather of Ubiquitous Computing (ubicomp),' Mark Weiser, has defined a smart environment as 'The physical world that is, richly and invisibly interwoven with sensors, actuators, displays and computational elements, embedded seamlessly in everyday objects of lives and has been connected through a continuous network'. The Internet has made a significant breakthrough in the course of achieving ubicomp's vision which facilitates individual devices to interconnect with any other device in the world.

The term Pervasive or Ubiquitous mean 'omnipresent or all-pervading.' Ubiquitous computing is a model in which the dispensation of information and data is connected with each movement or object as encountered. The primary tools that support ubiquitous or pervasive computing include Internet, Operating System (OS), sensors, microprocessors, networks, mobile code, and mobile protocols. While these supporting tools are a few of them, the technologies required to support

pervasive or ubiquitous computing is vast and varied. Ubiquitous or Pervasive Computing encompasses linking electronic devices, comprising embedding microprocessors to transfer information. Devices that use ubiquitous computing have uninterrupted accessibility and are totally connected.

Electronic product codes for specific or individual items have been developed for use on tiny tags fixed with radio transmitters. Japan's Universal ID Centre, for example, has executed a 128-bit addressing system for labelling distinct objects. This will allow the Ucode system to dispense '340 000 000 000 000 000 000 000 000 000 000 000' codes. Further, it authorizes a trillion tags to be apportioned every day for a trillion years, and despite have some left over, which seems infinite.

Addressing systems can be adequate for future connectivity needs, however, history teaches not to be certain of anything. With rapid increase in the growth of the Internet, the Internet needs to encompass further elements of the real world, and therefore, the idea has to be intricate. Simplicity will no longer be an option. Communications will increasingly grow ubiquitous in daily life, requiring increased identifying and addressing systems. The real world is mapped more closely to the Internet in increasing feature, with efforts to develop a structure for the Internet. In this framework, technological ubiquity and complication will initiate the future communication landscape.

1.1.5 Fundamental Components of IoT

IoT is not a technology. It is a system which permits modifications with certain degree of flexibility. The ability to exchange and make use of information (Interoperability) between the integrated systems of all the components prompts, a high level of complexity. The capability to handle the interfaces will determine the feature of IoT, to become a true Network-of-Networks.

The fundamental components that prove Internet of Things (IoT) as reality are:

- **Hardware** - Turns physical objects approachable and give the ability to recover data and react to instructions.
- **Software** - Facilitates the data collection, storage, manipulating, instructing, and processing.
- **Communication Infrastructure** - Most vital of all, is the communication infrastructure which includes protocols and technologies to support two physical objects to exchange data.

An IoT platform is about facilitating connectivity among things or devices. Its architecture comprises a software platform, analytic platform, or application development platform.

A true end-to-end IoT platform comprises eight significant architectural building blocks and they are as follows:

1. **Connectivity and Normalization**: Carries various protocols and data formats into single software interface to ensure exact data streaming and interface with every device.
2. **Device Management**: It ensures that, the connected 'things' works accurate and perfect through running patches. It also updates software and other applications running on the device or edge gateways.
3. **Database**: Mountable storage of device data generates the necessities for hybrid cloud-based databases to a fresh level, in terms of data volume, variety, veracity, and velocity.
4. **Processing and Action Management**: Gives life to data with rule-based event-action-triggers and enables accomplishment of 'smart' actions created on specific sensor data.
5. **Analytics**: Performs an array of compound analysis from basic data cluster along with in-depth system learning to projective analytics and excavate the most value out of the IoT data-stream.

6. **Visualization**: Facilitates humans to perceive patterns and witness trends from visualization dashboards where data is intensely depicted through stacked, line or pie charts as well as 2D or even 3D-models.
7. **Additional Tools**: Support IoT developers' prototype, examine and advertize IoT, using case creating platform, ecosystem applications to envisage, handle and monitor the connected devices.
8. **External Interfaces**: Communicates with third-party systems and rest of the IT-ecosystem through Software Development Kits (SDK), built-in Application Programming Interfaces (API), and gateways.

1.2 History of IoT

The Internet has progressed to link all signs of intellectual capacity around the world. A number of terminologies related to the concept and ability to connect each and everything and from anywhere, have made their presence. Experts had foresighted smart, interactive objects 45 years ago; before the global computer network was propelled. The IoT is a technological innovation that signifies the prospect of computing and communications. Users can access the information far and wide and ubiquitously from anywhere and from any device anytime. Connections will increase and generate a whole new dynamic system of networks, which comprise the IoT. The Internet has grown to link each sign of intelligence (that is, software) across the world, which includes wearables, Machine-to-Machine (M2M), ubiquitous computing, Radio Frequency Identification (RFID), context-aware computing, and the Web of Things. Considering the immense use of the IoT in the present world, it is important to know the pioneers of IoT technology and their projects that helped move it from its initial sights into today's most discussed topic. Figure 1.3 shows the Objects in IoT.

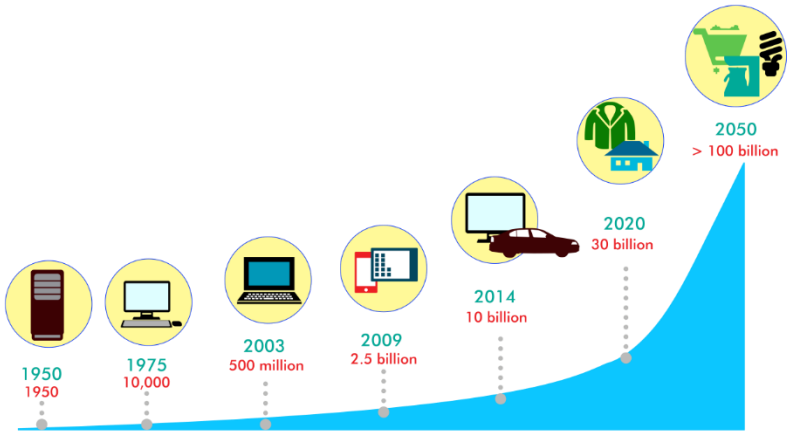


Figure 1.3: Depicts Objects of IoT

The term IoT was first conceived in 1999, by Kevin Ashton. Ashton is the Executive Director of the Auto-ID Centre at the Massachusetts Institute of Technology (MIT), Cambridge, USA. Auto-ID Labs, a research-oriented successor to the MIT Auto-ID Centre was incorporated in the year 1999. The other founders of Auto-ID are David Brock and Sanjay Sarma. They helped to improve the Electronic Product Code (EPC), a global Radio Frequency identification (RFID) system intended to replace the Universal Product Code (UPC) bar code.

On June 22, 2009, in the 'RFID Journal', Ashton quoted 'I could be wrong, but I am fairly sure the phrase 'Internet of Things' started life as the title of a presentation I made at Procter and Gamble (P&G) in 1999. Linking the new idea of Radio Frequency Identification (RFID) in P&G's supply chain to the then-red-hot topic of the Internet was more than just a good way to get executive attention. It summed up an important insight which is still often misunderstood'.

Ashton stated that 'In today's world computers and the Internet are entirely dependent on human beings for information.

Almost all the data available on the Internet were first reproduced and generated by humans by manually typing, or pressing a record button, or taking a digital picture, or scanning a bar code. With people not having adequate time, attention and a detail eye for perfection or accuracy, the generated data about things in the actual world is not of high standards. If computers could gather data without the help or intervention of human beings, it would be easy to detect and count everything and this would minimize wastage and lead to reduction in cost'. Ashton quoted in the 'RFID Journal': 'The IoT has the potential to change the world, just as the Internet did'.

Neil A. Gershenfeld, Professor at MIT and the director of MIT's Centre for Bits and Atom (CBA), a sister lab to the MIT Media Lab, too highlighted similar concerns like Ashton did. At the time of establishing the CBA, Gershenfeld published a book 'When Things Start to Think' in which he quoted,'In retrospect it looks such as the rapid growth of the World Wide Web may have been just the trigger charge that is now setting off the real explosion, as things start to use the Net'.

The pioneers of IoT technology and their projects that helped move it from its initial sights into today's world are listed in following Table 1.1.

Year	Pioneers	Projects
1926	Nicola Tesla	Tesla had prophesied that the present day smartphones would be the size of a shirt pocket that would be easy to carry and simple to handle.
1946		The two-way wrist-radio, which was worn by fictional characters Dick Tracy and his colleagues, made its first presence and became one of the comic strip's most identifiable icons.
1949	Norman Joseph Woodland	The Barcode was conceived when Woodland drew four lines in the sand on a Miami Beach.
1955	Edward Oakley Thorp	Known as the 'father of the wearable computer,' invented a tiny automatic device that is fitted on the body.
1960	Morton Leonard Heilig	Invented the Telesphere Mask, the first instance of a Head-Mounted Display (HMD), for the non-interactive medium, for example, films.
1966	Karl W Steinbuch	Developed Lernmatrix, an 'Associative-Memory-like Architecture,' and an initial implementation of Artificial Neural Networks (ANN).
1967	Herbert Upton	Created an 'analogue wearable computer' as a support for 'lip-reading.'
1969		The first message from ARPANET was transmitted. ARPANET became the technical base for the Internet.
1973	Mario Cardullo	Received the first patent for a passive, read-write RFID tag to mechanically categorize and detect tags attached to objects.
1974		For the first time, UPC label was used to ring up purchases at a supermarket.

1977	Carter C Collins	Invented a five pound wearable, which had a camera that could be fitted on the head. The system was tested as a visual prosthetic for the blind.
1981	Steve Mann	Mann developed a backpack-mounted 'wearable personal computer-imaging system and lighting kit.
1989	Sir Timothy John Berners-Lee	Designed the 'World Wide Web (WWW).' He executed the first successful message between a Hypertext Transfer Protocol (HTTP) client and server through the Internet.
1990	Olivetti	Developed an 'Active Badge' system, using electromagnetic signals to inform the person's location and record it in a central database.
1990	John Romkey	Invented an Internet toaster and connected the toaster to the Internet with TCP/IP networking, and controlled it with a Simple Networking Management Protocol Management Information Base (SNMP MIB).
1993	Steve Feiner, Blair MacIntyre, and Dorée Seligmann	Invented 'Knowledge-based Augmented Reality for Maintenance Assistance (KARMA)' – a Private Eye display over one eye, giving an analogous effect of the real world being viewed by both eyes.
1993	Thad Eugene Starner	Started using a specially-rigged computer and heads-up display as a wearable.
1994	Xerox EuroPARC's Mik Lamming and Mike Flynn	Exhibit the 'Forget-Me-Not,' a wearable device that communicates via wireless transmitters and records communications with people and devices, and storing the information in a database.
1994	B.N. Schilit and M.M. Theimer	Coined the term 'context-aware' for the first time in a Network article.
1995	Siemens	Sets up a department to develop and launch a Global System for Mobile communication (GSM) data module called 'M1,' for machine-to-machine industry applications.
1995	Steve Mann	Invented the first Wearable Wireless Webcam, the first example of Lifelogging.
1997	Paul Saffo	Projected that in the near-term technology expansions in sensors will help prepare people for the ensuing effect of nanotechnology.
1998	Mark Weiser	Constructed a water fountain outside his office whose flow and height emulated the volume and price trends of the stock market.
1999		Procter and Gamble's Kevin Ashton first mentioned IoT-denoting to the link between RFID technology in P and G's supply chain and the Internet.
2000	LG	Launched the world's first Internet refrigerator, the 'Internet Digital DIOS.'
2002	Ambient Orb created by David Rose	Showed that the Dow Jones 'personal finance and weather information' was centered on Internet data and changes its color based on the active strictures.
2003	BigBelly	A waste management company announced that it aims

		to transform its trash cans to double as wireless hotspots throughout New York City.
2003		• Projects such as, Cool Town, Internet0, and the Disappearing Computer Initiative seek to implement some of the concepts, and the IoT term begins to appear in book titles for the first time. • RFID is installed on an enormous scale by the US Department of Defense in their Savi program; by Wal-Mart as an incremental development in operational efficacy. • A new dimension had been added to the world of Information and Communication Technologies (ICTs): from anytime, anyplace connectivity for anyone, we will now have connectivity for anything. Connections will multiply and create an entirely new dynamic network of networks – the 'Internet of Things'.
2005	Interaction Design Institute Ivrea (IDII)	Created 'Arduino,' an inexpensive and simple to use 'single-board microcontroller,' for their students to use in developing interactive projects.
2005		The IoT reached new heights, when the UN's International Telecommunications Union ITU published its first report on the topic.
2005	Rafi Haladjian and Olivier Mevel	Nabaztag, a Wireless Fidelity (WiFi) device which is rabbit-shaped, collected information from the Internet and warned the users orally about vital messages and bulletin.
2006		Recognition by the European Union (EU) and the First European IoT conference is held.
2008		The IPSO alliance presently has over 50 member companies, including corporate giants such as Bosch, Cisco, Ericsson, Intel, SAP, Sun, Google, and Fujitsu.
2008		The Internet of Things was 'Born'.
2008		The U.S. National Intelligence Council warned that the IoT would be an interruptive technology by 2025.
2008		An array of IoT platforms (Pachube, Thingspeak), standards (6LoWPAN, Dash7) hardware and software (Contiki, TinyOS) was developed.
2010		The concept of IoT became popular. The same year news spread that Google's StreetView service had not only made 360 degree pictures but had also accumulated loads of data of people's WiFi networks.
2010		In May 2010, ZigBee Alliance and Internet Protocol version 6 (IPv6) Forum form strategic partnership with IPSO for speedy implementation of IP networked smart objects.
2011		Gartner, the market research company that designed the well-known 'hype-cycle for developing technologies' included an emerging phenomenon on their list: 'The Internet of Things.'

2011	Ericsson	In February 2011 the wireless firm envisaged that there would a whopping 50 billion IoT devices by 2020.
2011	Nest Labs	Introduced its 'Nest Learning Thermostat,' which uses sensor algorithms, machine learning and cloud computing to comprehend the home owner's behaviors and choices and fine-tunes the temperature by increasing or decreasing it accordingly.
2012	Google	Begins testing its Google Glass prototype, which is a pair of glasses with an optical HMD that exhibits information gathered wirelessly, according to the user's specification.
2012		On June 6, 2012, World IPv6 Launch Day was organized.
2012	Proteus Digital Health	In July 2012, US gets Food and Drug Administration (FDA) clearance for its ingestible medical device that wirelessly transmits information to the patient's vital signs to a patch on the skin, which then communicates the information to a mobile phone.
2012		Internet of Things grows rapidly on social networks such as LinkedIn and the UK's Technology Strategy Board networking platform.
2013		• In October 2013, International Data Conference (IDC), Massachusetts, US, published a report stating that the Internet of Things would be an $8.9 trillion market in 2020. • Cisco, IBM, Ericsson created huge scholastic and marketing creativities on the topic. • Arduino and other hardware platforms mature and make the IoT accessible to 'Do It Yourself consumers (DIY'ers)' taking interest in the topic. • Acquisitions and Venture Capital (VC) investment in the space including the IoT platform Pachube being acquired, IoT security company Mocano raising a round of funding, and other VC's taking notice of the industry.
2013		In December 2013, Chip maker Qualcomm, along with other tech firms, forms the' All-Seen Alliance (ASS)', which is intended to develop an open framework to enable the Internet of Things.
2014		• In April 2014, Google Glass goes on sale to the general public for a hefty $1,500. • In July 2014, Chip maker Intel and other high tech firms set up a rival group to 'All-Seen Alliance (ASS)' to promote an open framework for the IoT called the 'Open Interconnect Consortium (OIC).' • The term IoT and its familiarity spread around mass market, when in January 2014 Google declared to purchase Nest for $3.2bn. At the time, Consumer Electronics Show (CES) in Las Vegas was held, with IoT being the dominant theme and topic.
2015		The Internet is reaching every corner of the world. By

		2020, the Internet is expected to have 7.6 billion users. Microsoft, Apple, Oracle, HP, Google, GE and many others have some sort of IoT platform established. There are also thousands of new IoT products being introduced each year and by the end of 2015 there would be 4.9 billion things connected to the Internet. With so many IoT related devices entering the market and so many different platforms that are available to choose from Tesla's grand vision of turning the Earth into a huge brain will be accomplished.

Table 1.1: Lists the Pioneers of IoT and their Projects

Figure 1.4 shows the interest in IoT over time.

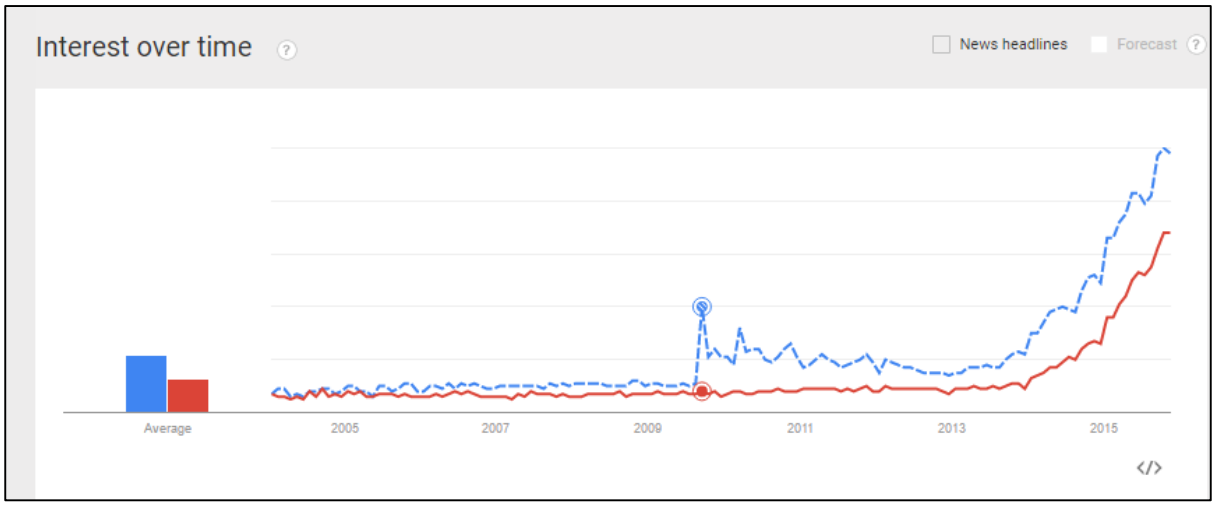


Figure 1.4: Depicts Interest in IoT Over Time

1.3 Objects or Things in the IoT

IoT is expanding enormously, from a miniature device to colossal machines. There are various smart devices that use 'wireless technology' to transfer information to each other and among humans. The quantum of objects in the 'IoT world' is rising at a rapid pace, from two billion objects in 2006 to a projected two hundred billion objects by 2020. This amounts to a ratio of 26 smart devices for each person around the world.

Majority of IoT smart devices have already occupied a significant place in the commercial front, such as manufacturing sectors, businesses, and in healthcare industry. However, our homes are yet to perceive and experience the technology of IoT smart devices.

IoT smart devices contribute to storing and sharing of critical data that is of importance to the industrial sector. For example, data required to check inventory, operate machines, increase productivity, save costs, and even secure lives. It is estimated that by the year 2025, the universal worth of IoT technology would probably be as much as USD 6.2 trillion; with major contributions coming from healthcare (USD 2.5 trillion) and manufacturing sectors (USD 2.3 trillion).

Figure 1.5 shows some of the Objects in IoT.



Figure 1.5: Depicts the Objects in IoT

According to sources, the percentage of IoT smart devices used by various sectors is shown in chart 1.1.

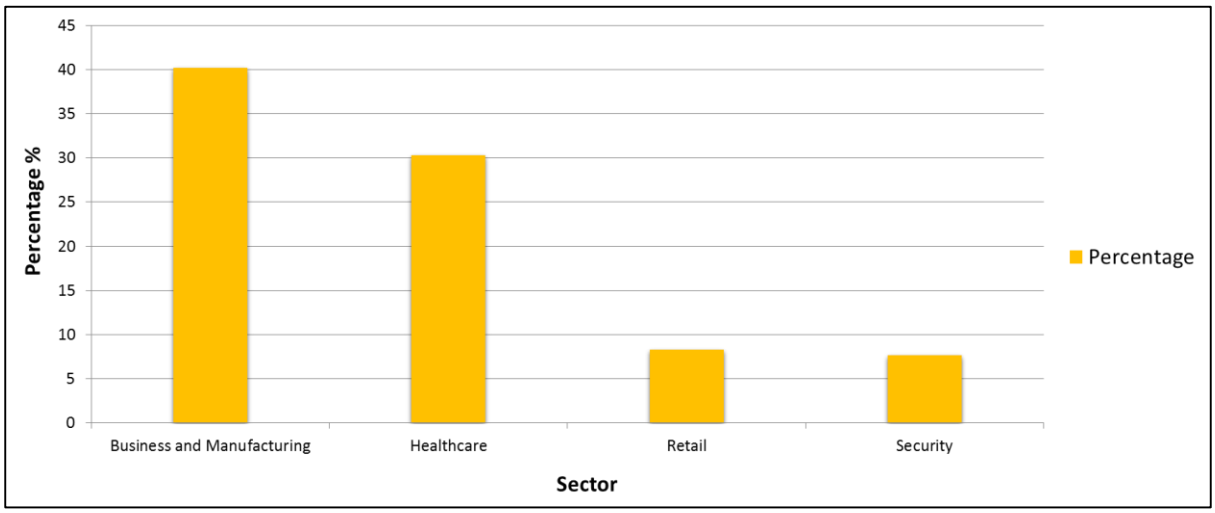


Chart 1.1: Percentage of IoT Smart Devices used in Various Sectors

Various Sectors that use IoT smart devices are as follows:

- **Business and Manufacturing Sector** - It includes Real-time analytics of supply chains and equipment, and robotic machinery.
- **Healthcare Sector** – It includes Portable health monitoring, automated record keeping, and pharmaceutical safety measure.
- **Retail Sector** – It includes Inventory tracking, smartphone purchase, and unspecified analytics of consumer preferences.
- **Security Sector** - It includes Biometric and facial recognition locks, and remote sensors.

IoT includes wide range of connected objects. Corporate Giant chip maker Intel, quoted, 'Computers smaller than a grain of sand can be sprayed or injected almost anywhere to measure chemicals in the soil or to diagnose problems in the human body'.

1.3.1 Smart Objects

A thing or object in the context of IoT is an entity or physical object that has a unique identifier, an embedded system and the ability to transfer data over a network. The popularity of wireless

technology has initiated miniaturization of computer components and developing them economically. The extensive range of the Internet addresses has led to capacity that can be developed for things beyond imagination. In addition to computing devices such as PCs, notebooks, smartphones or tablets, the list of impending objects or things is limitless. Some examples of the objects or things of IoT are as follows:

- **Smart Power Meter**: Smart meter is an automatic device that logs consumption of 'electric energy in interims of an hour or less and transfers that data back to the energy supplier for tracking and billing. This will guarantee more detailed and precise energy bills. Smart meters support two-way interaction between the meter and the energy provider. Smart meters can collect information for distant reporting. The first smart meter for both electricity and gas was installed in early 2005.
- **Smartphones:** Smartphone is the key IoT device. It makes lives smarter in many ways, such as, banking, bill payments, fitness and health monitoring, and personal computing. It made connectivity easy and users can operate other IoT devices, such as, smart TVs and digital locks with smartphones.
- **Security and Safety**: Digital locks are the next advanced technology in security. Smartphones can be used to lock and unlock doors remotely and business owners can alter key ciphers swiftly to allow or limit access to employees and guests. Security cameras can monitor and transmit the data to smartphones.
- **Fitness Devices**: There are many devices available for personal fitness, such as, smart weighing machines that can connect with smartphones, heart-rate monitors, and fitness bands.
- **Smart TVs**: Televisions are in existence for past few decades. Television has evolved at a greater speed, since its inception. Initially, it was a Black and White television, and then the color television emerged. The flat TVs, LCD TVs, LED TVs are examples of technological advancement. Technology has reached new heights in the history of TVs - The Smart-TV, has several feature, such as, connecting to Internet, share videos, store channel programs, personal computing, online games, and so on.
- **Smart Bulbs**: This device can be operated by a smartphone, it automatically turns ON and OFF based on the predefined activities or the room lighting conditions and can change its color according to the mood or weather.
- **Personal Entertainment**: Devices in entertainment world are massively improved; the gadgets such as portable speakers can be connected through Wi-Fi or Bluetooth.
- **House Cleaning Robots**: These robots can automatically clean our homes and can be controlled using smartphone. Many companies such as LG are selling house cleaning robots in the market. In near future every house-hold will have such robot/device to clean house.

1.3.2 The Internet Evolution

It is anticipated that by the year 2020 human intelligence and machine intelligence will merge. There will be no necessity to use a keyboard and mouse to control computers. Instead, users will open documents and surf the Web using their brain waves.

Researchers are working on finding ways to read and harness human brain waves to operate computers, television sets, and cell phones. Sensors implanted in human brain would harness the brain waves. Users will stop depending on a computer interface or operate an interface with their fingers. Instead, they will operate various devices with their brains. Researchers are gaining the ability to build brain sensing technology into a head set that could be used to operate a computer. The next step is development of a tiny, far less cumbersome sensor that could be implanted inside the brain.

In future, networked robots will have the ability to absorb from each other and work in teams to increase productivity and solve scientific problems.

1.3.3 Advent of Internet Era

The 'Internet' is an informal term for global communication network of computers. The Internet is used to send data rapidly between computers everywhere in the world. It has innumerable smaller domestic, academic, commercial, and government systems and Websites; which together carry different types of information and services.

Earlier televisions, phones, bulbs, coffee makers, homes, machines, and each thing performed only specific function that they were designed or invented for. The advent of the Internet technology made objects or things smarter such as smart-TVs, smartphones, smart bulbs, smart homes and every aspect of our daily life are getting smarter each day.
Figure 1.6 shows the model of the Internet.



Figure 1.6: Depicts the Model of the Internet

IoT links various devices and is very compact, connecting billions of objects. An Internet, IP, or TCP/IP built model stands at the centre of the IoT. It is a conceivable working solution to hide the ever-increasing heterogeneity of interacting technologies (the various technologies being diverse but not comparable with each other) and communication systems in the ubiquitous environment intended. The IP (method by which data is sent from one computer to another) might not back the resource constraint and scalability of the system. IP or the Internet will undoubtedly support activities aiming at designing new devices in IoT applications.
IoT research development will expectantly come with an innovative IP model and structural design. This will provide solutions to issues arising from scalability, resource constraint, and conception of new applications.

The model and design comes under the post-IP or future Internet, where several research projects are developing a new transmission model and design that familiarize the requirements of a specified system. These research projects are currently supported by the all-IP network; the packet-switching TCP/IP model has replaced the classical telecom circuit-switching model. The Internet presently is the widespread model in telecommunications to provide different services. The current TCP/IP model might be compatible with the emerging post-IP or future Internet model. Formulating policies for IoT services and network will necessitate revising of the basic concepts related to addressing, routing, scaling, and guaranteeing quality of service, security, mobility, and so on.

1.4 IoT in Cloud Computing

Cloud computing is a type of Web-based computing that provides applications and services to users on demand. Cloud computing involves provisioning of shared processing resources and data to users on request. Cloud computing resources can be provisioned by the users without requiring interactions with the cloud service providers. Cloud computing enables pervasive, on-request access to a shared group of computing resources that can be configured (such as, networks, services, applications, servers, and storage), which can be swiftly provisioned (arranged) and delivered with least management effort. Cloud computing and storage solutions offer users and organizations with several abilities to store, monitor, process, and control their data in external data centers. Cloud computing relies on sharing of resources to attain consistency and cost advantages.

Cloud Computing resources can be accessed over the network using standard access mechanism that offer platform-independent access through the use of various user platforms. Cloud Computing resources are pooled to serve (to be served by the same physical hardware) multiple users using multi-tenancy. Users are also assigned virtual resources that can run on top of the physical resources. The virtual infrastructure for utility computing integrating applications, storage devices, monitoring devices, visualization platforms, analytics tools, and client delivery are provided in cloud computing.

Cloud computing offers the utility-based model that will enable businesses and users to access applications on demand anytime, anywhere, and anyplace. Cloud computing also enables enterprises to get their applications active and running more rapidly, with better manageability and less upkeep, and assists Information Technology (IT) to rapidly order resources to meet inconsistent and volatile business demand. Cloud providers normally use a ‘pay as you go’ (paying for a service before it is used) model. This leads to erratically high cost if managers do not familiarize to the cloud pricing model.

Figure 1.7 shows a model of Cloud computing.

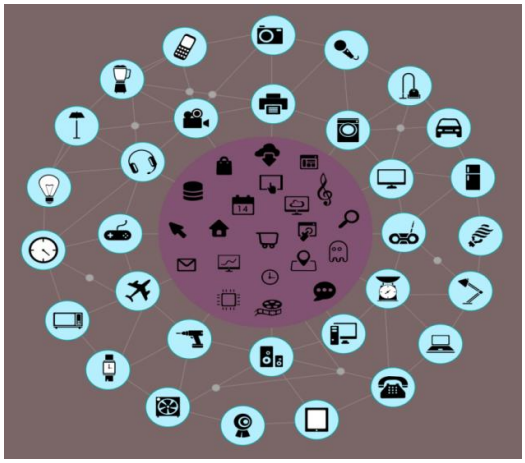


Figure 1.7: Depicts Model of Cloud Computing

1.5 IoT in Big-data

Big-data is defined as ‘a term for data sets that are so large or complex that traditional data processing applications are inadequate. Big-data comprise massive volume of both structured and unstructured data that is so large that it is difficult to process using traditional database and software techniques’. The volume of data is too large and it moves too rapidly and exceeds the present processing capacity. The difficulties in processing the such huge volume of data is in the analysis, data encapsulation, data management, data search, data sharing, data storage, data

transfer, data visualization, data querying, updating, and information privacy. Precision in Big-data can lead to confident decision making and better decisions results in greater operational efficiency, reduction in cost, and reduced risk. Despite these difficulties, Big-data has the capacity to assist companies' better operations and take more intelligent decisions. The data can help a company to gain from increase in revenues, retain customer base, and improve operations. Big-data and the IoT work in unification. Big-data is the key product of device inter-connectivity and enables precise targeting.

The IoT, with the help of Big-data, transforms the industries, companies, and governments by opening up a new era of economic growth and competitiveness. IoT and Big-data can be used to better operations. It helps to control where data is produced across a wide range of 'vertical markets' (a market comprising all the potential purchasers in a particular occupation or industry) and exhibits the main vertical markets where Big-data is compiled. The vertical markets are not restricted to agriculture, electricity, forestry or water treatment, but each type of manufacturing facility uses it. IoT and Big-data are put together into practice to have less downtime, predictive health monitoring, improve quality, higher quantity, lower reject rates, more efficient use of labor, improved safety and enable mass customization of manufacturing. Related vertical industry operations can be improved by the intelligent implementation of the IoT and Big-data. These operational developments results in better products, less downtime, increased quantity, and lower the costs.

Big-data is a group of data series with dimensions over and above the ability of commonly used software tools to process, within an acceptable lapsed time. Examples of Big-data include:

- Extensive E-commerce
- Universal financial data
- Internet script and documents
- RFID
- Web records
- Medical records
- Photography archives such as Flicr
- Video archives such as YouTube
- Big Science
- Sensor networks
- Social networks and social data such as Facebook
- Internet search indexing such as Google
- Call detail records
- Astronomy
- Atmospheric science
- Genomics
- Biogeochemical research data
- Biological study data

1.6 Summary

- Internet of Things (IoT), is the way to a smart-world with ubiquitous computing and networking.
- IoT-based services concentrate on offering more automation of many tasks around people and allied objects so as to build a smart world not only in manufacturing units, also in office, at home and across the world.
- Quality of our life is improving in numerous ways with smart IoT devices; whether it is banking or shopping it can be done fairly with a single click.
- Internet of Things (IoT) is not a technology, however, a system that permits for modifications with a certain degree of flexibility.
- IoT includes wide range of connected objects computers smaller than a grain of sand can be sprayed or injected almost all over to measure chemicals in the soil or to sense problems in the human body.
- Combination of IoT along with cloud computing, support sensing services and commanding process of sensing data stream.
- Big-data functions vary for each single IoT device it is connected, on the other hand chiefly it captures and stores all the incoming data. This information will be examined in one point center to improve the performance and process.

1.7 Check Your Progress

1. IoT based services concentrate more on offering _____.

    a. Automation
    b. Free Subscription
    c. Free Internet
    d. Free GPS

2. What is the actual purpose of IoT?

    a. Making things secure
    b. Making things smarter
    c. Making energy efficient
    d. To provide Internet

3. What are the fundamental components of IoT?

    a. Hardware
    b. Software
    c. Communication Infrastructure
    d. All of these

4. Who introduced the term Internet of things?

    a. Tim Berners-Lee
    b. Steve Jobs
    c. Kevin Ashton
    d. Nicola Tesla

5. What is Cloud computing?

    a. Wireless computing
    b. Internet based computing
    c. Peer-to-Peer computing
    d. Browser-free computing

1.7.1 Answers

1	**A**
2	B
3	D
4	C
5	B

SESSION 2 - IOT IN THE REAL WORLD

Session Objectives:

This session explains the role of IoT in the real world. This session also provides a comprehensive study on IoT versus the smart world and its role across various domains that are spanning many business industries.

At the end of this session, students will be able to:

- Explain the concept of IoT in real world
- List down the features of IoT in smart world
- Explain the role of IoT in the growth of diverse sectors

2.1 Introduction

IoT is a network of devices, linked over cloud-computing networks that leverage machine-to-machine communication. This allows its devices to transmit and coordinate data in real-time. IoT adds the word 'smart' in each of its application objects such as smart homes, smart cars, smart cities, smart watches, smart energy, smart payments, and so on. IoT aims in provisioning real-world services, which is delivered by smart objects that are related to this physical world.

IoT manipulates information processing and knowledge engineering methods to observe and measure data that are acquired from physical objects using Internet data. It brings the creativity between 'smart' and connected things. IoT implements sensor that is unique to its application object to make use of that data in the real time scenario.

The impact of IoT will be more reflective in the consumer products comprising embedded intelligence.

The IoT represents an idea in which the Internet encompasses the real world involving everyday objects. The IoT vision is built on the certainty that the steady progresses in microelectronics, communications, and IT will endure in the anticipatable future. Due to their miniature size, declining cost and lesser energy consumption, communications modules, processors, and other electronic components are being gradually embedded or implanted into daily life objects.

Smart objects using sensors through embedded communications and built-in networking abilities, helps to identify another smart object to interconnect with each other and access Internet services, which in turn networks with people.

2.2 Home Automation

Home automation system is described as a residential building or home in which all devices or appliances are connected to a central hub and is remotely controlled with mobile devices through Internet. Home automation provide users the benefits of increased comfort, efficient control of

building's system, lower energy consumption, decrease in overhead costs, and increase in life span of utilities.

Many companies are designing homes with automation technology, which enable users to control all home appliances and devices, functions and features either automatically or remotely. For example, users when travelling can use their mobile or tablet to remotely control their security system, regulate thermostat, lightings, program an entertainment system, and perform several other tasks.

Home automation system depends on open source home automation platforms, which support large number of devices and integrations that communicate information of the state of home to the users. Home automation system uses large number of sensors from which users manage, access, and retrieve information through smart devices, such as, mobile phones, touch screens, and so on. These sensors includes motion and occupancy sensors, light sensors, temperature and humidity sensors, fire alarm sensors, gas leak sensors, home security sensors, and so on.

2.2.1 Smart Home

Smart home is defined as a residence wherein all home appliances' features, controls, and functions, such as, lighting, air conditioning, heating, TV, entertainment, audio and video systems, and security systems communicate with each other and can be remotely controlled by a program. A smart home is designed with advanced automation systems to enable users to monitor and control the building's functions effectively and efficiently.

Figure 2.1 shows the smart home.



Figure 2.1: Depicts a Smart Home

Smart homes provide users with intelligent response and information by controlling many features of a home. For example, unlocking door automatically, set the room lightings according to the ambience or users' preferences, controlling the security systems, and so on.

Nowadays builders or people buying new homes are installing additional wiring and controls that facilitate operation of sophisticated home automation systems. Thus, if these homes needs to be upgraded with smart home technologies then, additional expenses for routing wires and placing sensors can be saved. Home automation systems are not restricted to a set of finite possibilities. Features can be added whenever the user prefers or when the users' budget allows.

Most features require users to operate a simple remote controller or an application on their smartphone. Once the users install it, there will be no additional costs related to it, unless it needs Internet to operate it remotely. The variety of smart home technologies is expanding with

advancement in computer controls and sensors. Smart homes make life stress-free for users, provide users with comfort and convenience, and contribute to efficiency and savings in energy.

Electricity bills decreases when lights are automatically turned OFF in vacant rooms and room temperature can be regulated based on the users preference. Smart home technology assures remarkable benefits for elderly people residing alone.

Few of the examples are as follows:

- Special alerts are send if any device is accidentally left on for a long period of time
- Alerts will be sent if sensors sense the presence of any harmful gases caused due to any leak
- Send an alert when it is time for the user's medication
- Systems with motion sensor triggered lights ensure elderly people do not struggle while walking in the rooms in the dark
- Surveillance and security system inside the house and around the exterior of the house helps in preventing burglary
- Alerts are sent to other members of the house or neighbors when elderly people accidentally falls in their home.

2.2.2 Smart Lighting

Smart lighting is a technology that aims energy efficiency by automated controls in accordance to various factors.

Few of the factors for automated control are as follows:

- Occupancy
- Availability of daylight

Smart lighting comprises Solid-State-Lighting (SSL), such as, Light-Emitting Diodes (LEDs), Organic Light-Emitting Diodes (OLEDs), or light-emitting polymers, and IP-enabled lights.

The primary feature of smart lighting is that it can be controlled from anywhere and at any time without having to depend on a wall-mounted switch. It can be controlled wirelessly through applications on smartphones or tablets. For example, shutting down the lights in the night, dimming a room to watch TV, or turning OFF lights that kids left on in their rooms, and so on. The sensors in smart lightings can be configured to adapt lightings based on users' preference for luxury, or the room's temperature, or occupancy, and so on. For example, color adjustable lights are used to enhance and change the look of a room according to users' preferences.

Smart lighting can result in energy saving in different and unique ways, such as, sensing the human movements and their ambience and controlling lightings accordingly. Smart lightings can also be programmed to automatically dim the lights, which enable energy efficiency, without users noticing or feeling the change in light output. Smart lighting enables lighting to be a vital part of a room's design and architecture, thus, adding onto the splendour of a home.

Figure 2.2 shows smart lighting control on a mobile device.



Figure 2.2: Smart Lighting Control on a Mobile Device

2.2.3 Smart Appliances

Handling and managing modern appliances, such as, air-conditioner, music systems, washer/dryers, refrigerators, TVs, and so on can be laborious, as each appliance is designed with unique controls or remotes.

A smart appliance is designed to make functions simpler, faster, and more energy-efficient. Smart appliances comprise thermostats, washing machines, microwaves, dryers, heaters, and refrigerators that are connected to Internet through WiFi connection and can be accessed and controlled remotely from mobiles, tablets, and computers.

Smart home technology provides a central automation system for smart appliances making them effectively smarter. Smart home appliances are set, so that, they can be controlled centrally or can be set to communicate with one another to make decisions for the owner. For example, Smart thermostats allows regulating the temperature remotely and study users' preference. Smart refrigerators can view and capture images of food items stored in the fridge and send it to users on their smartphones. Smart oven can be programmed to heat or cook a meal, when activated by the user through Internet. Smart washing machine enables users to pre-set washing itinerary through mobile devices. Home automation technology connects all appliances to a central system or hub, from where they can be controlled remotely.

Other household appliances, such as, lighting, TV, sound systems, garage doors, gates, ventilation, locks, shades, and hot water systems have also adopted smart technology. Sensors, can detect water and gas leaks and will cut power if there is a short circuit. Smart home appliances are designed to reduce power and energy usage that essentially help in reducing expenses.

Figure 2.3 shows the smart appliances used in recent homes.

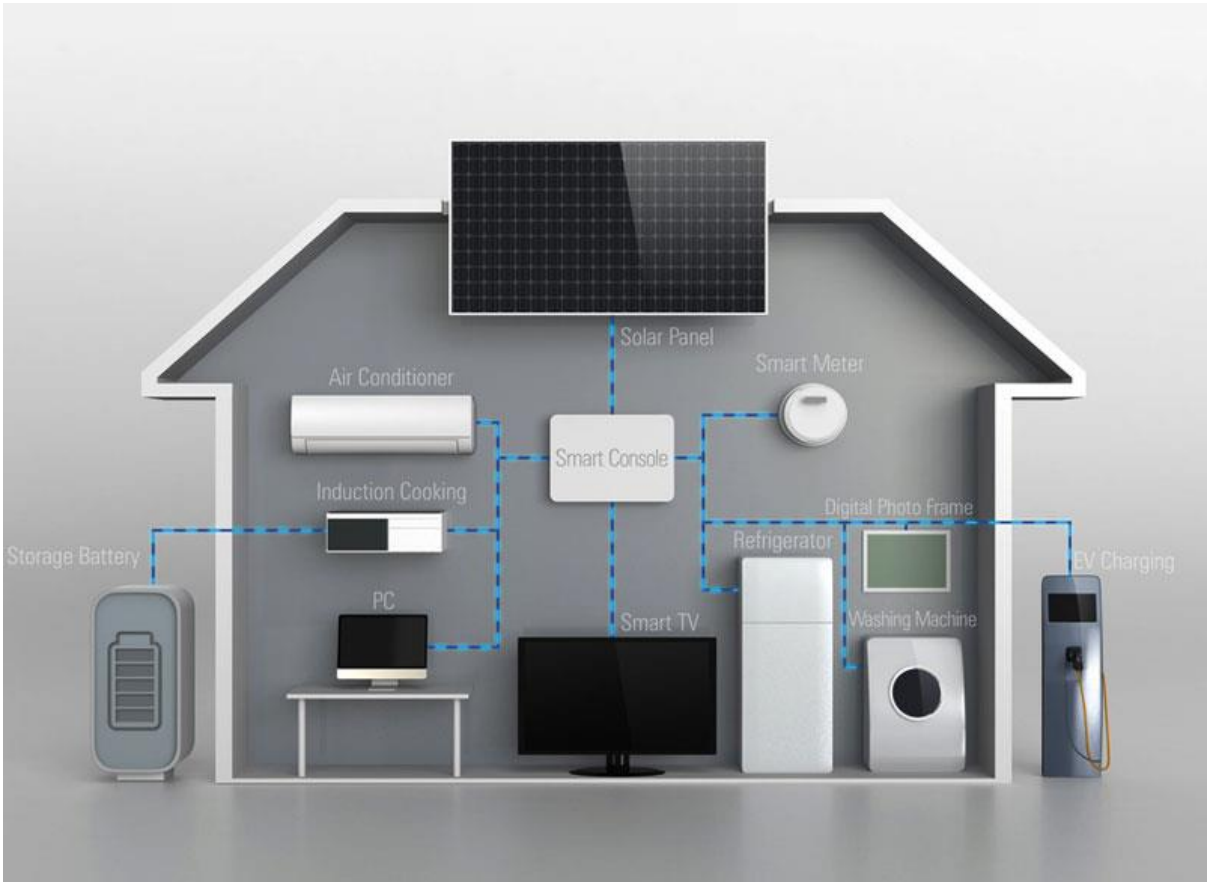


Figure 2.3: Smart Appliances

2.2.4 Home Intrusion Detection System

Home intrusion detection model is a system designed to detect, sense and identify intrusion, or unauthorized entry, or suspicious movements within the house, if nobody is home. The system use Passive Infrared (PIR) sensors, motion sensors, ultrasonic sensors, door sensors, infrared light beams, glass break sensor, inertia sensors, and security cameras to sense intrusions and send alerts promptly to users. The PIR motion detector is the most common sensors found in homes and are inexpensive and highly reliable.

PIR sensors do not sense movement but they sense sudden changes in temperature at a given point. When an intruder walks in front of the sensor, there will be sudden raise in the room temperature that will automatically elevate the body temperature at that point of intruding and comes back to normal after some time.

This rapid change in temperature triggers the detection and send alerts to users. Alerts can be sent through SMS or as an E-mail to the users. Advanced systems also send image or video alerts in the form of E-mail attachment to users or local law enforcement if there is unauthorized entry.

Audio and video identification systems use cameras and microphones to record image snapshots, audio frequencies, or video signals. In cases of any intrusion sensed or detected, cameras will automatically start recording and the home server will send alerts to owners, neighbors, and nearby police stations. The information transmitted to law enforcement is recorded to an event file and can

be later used as prosecution evidence. Alarms with audio, video, or combination of both audio and video identification technology offer property owners, security companies, and police more reliable information to measure the threat level of intrusions or break-ins.

Figure 2.4 displays Intrusion Detection Systems.



Figure 2.4: Intrusion Detection Systems

2.2.5 Smart Smoke and Gas Detectors

Smoke detectors and Gas detectors can detect the presence of fire and harmful gases in homes and alert home owners in the event of such incidents. Smart smoke and gas detectors are connected to WiFi, so users can control and monitor battery status, receive alerts, or warnings of smoke detection through mobile devices, even outside the home.

Alerts by SMS or E-mail, or colored alarm codes, enable users with hearing imparity to also install smart smoke detectors in their home. Smart gas detector is easy to mount and monitors the presence of harmful gases in home. Alerts are sent through sms or E-mail to users' mobile devices or through a human voice, of the presence of fire or likely fire hazard.

Voice alerts can be programmed to warn users in which room smoke or excess harmful gas is found. For homes that already have smoke and gas detectors, retrofitting their smoke alarms may be a solution, since, the cost of changing or replacing them with smart detectors can be a constraining.

Figure 2.5 shows the Smoke and Gas Detectors.



Figure 2.5: Smoke and Gas Detectors

2.3 Smart City

A smart city means integrating IT and communication networks in a secure manner to manage infrastructure comprising local departments, schools, waste management transportation, information systems, libraries, hospitals, water supply networks, power plants, law enforcement, energy, and other community services. Smart city enhances quality of life and urban development using technology.

Information and Communication Technology (ICT) improves quality, interactivity, and performance of urban facilities, decrease costs and resource usage, and develops the efficiency of services and meet resident's requirement. The use of wireless sensors enables monitoring infrastructures in real-time. Information gathered from such sensors are processed and analyzed to tackle inefficiency and enable a better quality of life.

Some of the ways to efficiently tackle the day-to-day life of Smart cities are as follows:

- Smart Parking
- Smart Lighting for Roads
- Smart Roads
- Structural Health Monitoring
- Surveillance
- Emergency Response

2.3.1 Smart Parking

The rapid surge in the population among cities directly has influenced the traffic density in the same proportion. The increase in number of vehicles has resulted in traffic congestion and difficulties in finding parking space. Smart parking system enables drivers to find parking space stress-free, and conveniently, thus, saving time, fuel and related costs.

Smart parking systems enable cities to manage their parking system more efficiently and reduce traffic jam and accidents. Smart parking use wireless sensors that are installed in each parking slot, to detect the number of vacant parking spaces.

This information is sent to central Web-server using Wi-Fi networks in real-time and drivers can retrieve this information via tablets, smartphones, and in-car navigation systems through smart parking applications installed in their devices.

Smart parking reduces traffic density, thereby reducing pollution and creating a greener city.

Figure 2.6 shows the smart parking of a city.

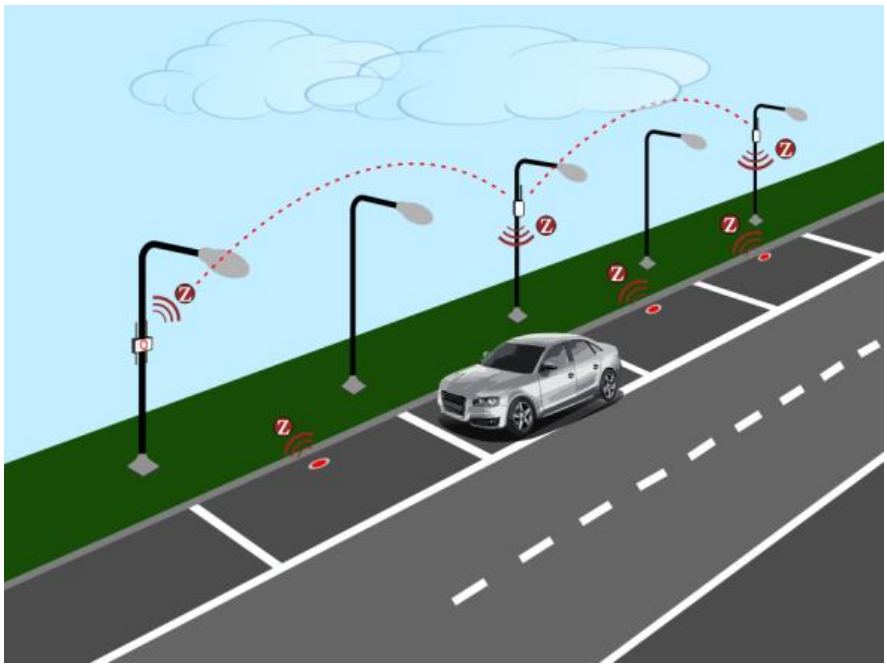


Figure 2.6: Smart Parking

2.3.2 Smart Lighting for Roads

Energy can be saved to a large extent with smart lighting systems for roads, buildings, and parks. Smart lighting controls lighting dynamically and adapts lighting to ambient conditions. Smart lights can be monitored in real-time and configured to set lighting schedules and intensity. Lighting can also be configured according to different weather conditions, such as, summer time, rainy season, poor visibility due to heavy mist, and so on.

Smart Lighting technology can also be used on Smart Roads by fitting the roads with power-saving lights that slowly illuminate as vehicles approach then switch OFF automatically after vehicles pass. A photo-luminescent powder mixed with road paint is used as road markings, which will illuminate the runway during the night, thus, facilitating safe driving. Smart lighting use wireless sensors that work in unison with other lighting connected to the same network to adjust lighting as per various conditions.

Figure 2.7 shows the Smart Lighting procedure.

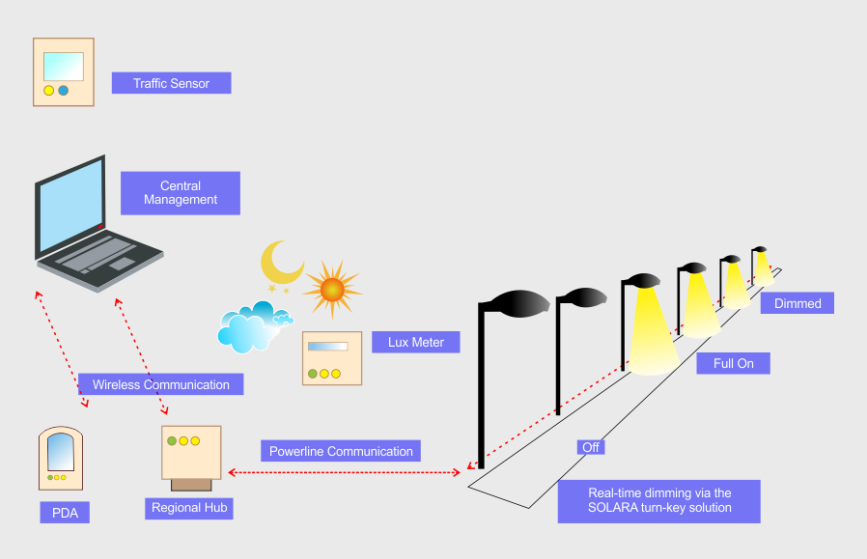


Figure 2.7: Smart Lighting for Roads

2.3.3 Smart Roads

Smart roads help drivers by providing information on condition of road surfaces, such as, potholes on the road surfaces, weather conditions related to road safety, and travel time, thus, making roads safer for driving and reducing traffic jams. Smart roads also enable monitoring damages and cracks, and monitoring systems in tunnels for air flow and visibility. Tunnels are monitored for presence of pollutants, such as, CO, CO2, NO2, O2, SH2, and PM-10 in the quality of air inside tunnels. Smart roads use wireless sensor networks and information from sensors are transferred to a central Web-server and distributed to drivers who subscribe to smart road applications.

Sensors and smart applications for monitoring weather, measures humidity, temperature, and other comparable parameters that enables driving safely in highways and avert road accidents. During winter, when road are filled with snow and ice, smart roads use solar power to facilitate clearing streets of ice and snow by melting it away. Smart road applications provide drivers with a consistent view of the road situation, thus, averting potential dangers that can arise from road accidents.

2.3.4 Structural Health Monitoring

The process of implementing a damage detection and categorization plan for engineering structures is described as Structural Health Monitoring (SHM). The SHM involves observing the structure over a period, from samples derived from sensors that are deployed on these structures. The sensors provide vibration levels and damage-sensitive information that helps in determining by statistical analysis, the present state of the structure's health. The result from such analysis gives periodic information related to the functional capacity of the structure, in the presence of unavoidable aging and degradation resulting from operating environments.

In the event of natural calamities, SHM enables instantaneous screening of the condition of the structure, which provides reliable information regarding the strength of the structure. SHM concept is widely applied to various infrastructures, such as bridges, skyscrapers, and monuments. By examining the information, it is possible to:

- Detect the existence of damages on the structure
- Locate the damages

- Identifying the types of damages
- Quantifying the severity of the damages
- Calculate the remaining life span of the structure

2.3.5 Surveillance

From surveillance perspective, smart cities monitor public through sensors placed in various places around the city. Data related to factors influencing urban living are received from such sensors. This data is gathered and analyzed by local authorities and governments to obtain information related to the challenges faced by the city in areas, such as, traffic management, crime prevention, waste reduction, and energy use.

This enables government towards better urban planning and in providing better services to the local population. Smart city surveillance technology is effective for efficient enforcement of law for prevention of crime; the development of essential infrastructure systems; optimization of transportation services; and offering local government services, such as, education, paramedics, healthcare, postal services, water supply, and so on through e-Governance portals.

For the purpose of city, wide surveillance video cameras are designed and placed in key areas in cities. The data from surveillance cameras can be collected in the back-end and can be developed to search for patterns or specific events for effective and efficient monitoring of public.

2.3.6 Emergency Response

IoT systems and sensors can be used for monitoring essential infrastructure in cities, such as buildings, gas and water pipelines, public transport, and power sub stations. Information from sensors on vital infrastructure are gathered, analyzed, and distributed to various government departments for monitoring vital infrastructure effectively.

In case of damage to city's gas and water pipelines alerts are sent to the concerned public works department of the potential hazards, thereby minimizing effects on the essential infrastructure.

In case of fire hazards alerts in the form of audio and video feeds are sent to the fire department, where information is analyzed in real-time to detect the intensity of the danger.

Emergency Response alerts can be in the form of alerts sent to public, redirecting of traffic, evacuations of affected areas, and so on. In case of sudden road accidents or an emergency, IoT traffic management system helps in minimizing disruption of regular traffic by adapting traffic light changes in accordance to the emergency, changing driving strategies of drivers depending on the criticality of the situation, reducing delay of emergency vehicles, such as, ambulances and police cars, and applying vital security controls.

2.4 Smart Environment

In Smart Environment, the demands placed on the environment can be met without decreasing its capacity to enable the people to live healthy, now and in the future. Smart Environment is defined for viable development and the three vital features for this are renewable energy, sanitation, water, and waste water management.

Air pollution affects human health and the environment and air quality surpasses emission limits for several air pollutants that pose grave health issues. Environmental noise is also a serious environmental problem. It can have physiological effects on people, restricting their basic activities, such as rest, study, sleep, and communication.

2.4.1 Weather Monitoring

Advanced weather monitoring system equipped with wireless sensors enables to monitor the environmental conditions. The wireless sensors constantly monitor the real-time weather conditions. Sensors constantly measure the climatic changes and send the data to a central hub or back-end. Weather monitoring system collect information from sensors related to temperature, air relative humidity, wind, atmospheric pressure, rainfall, light intensity, and so on.

The data collected enables to send weather alerts to the subscribed users from such applications installed in their smart devices. This allow users to take decisions on their daily schedule based on the weather conditions. Figure 2.8 shows the Weather Monitoring.

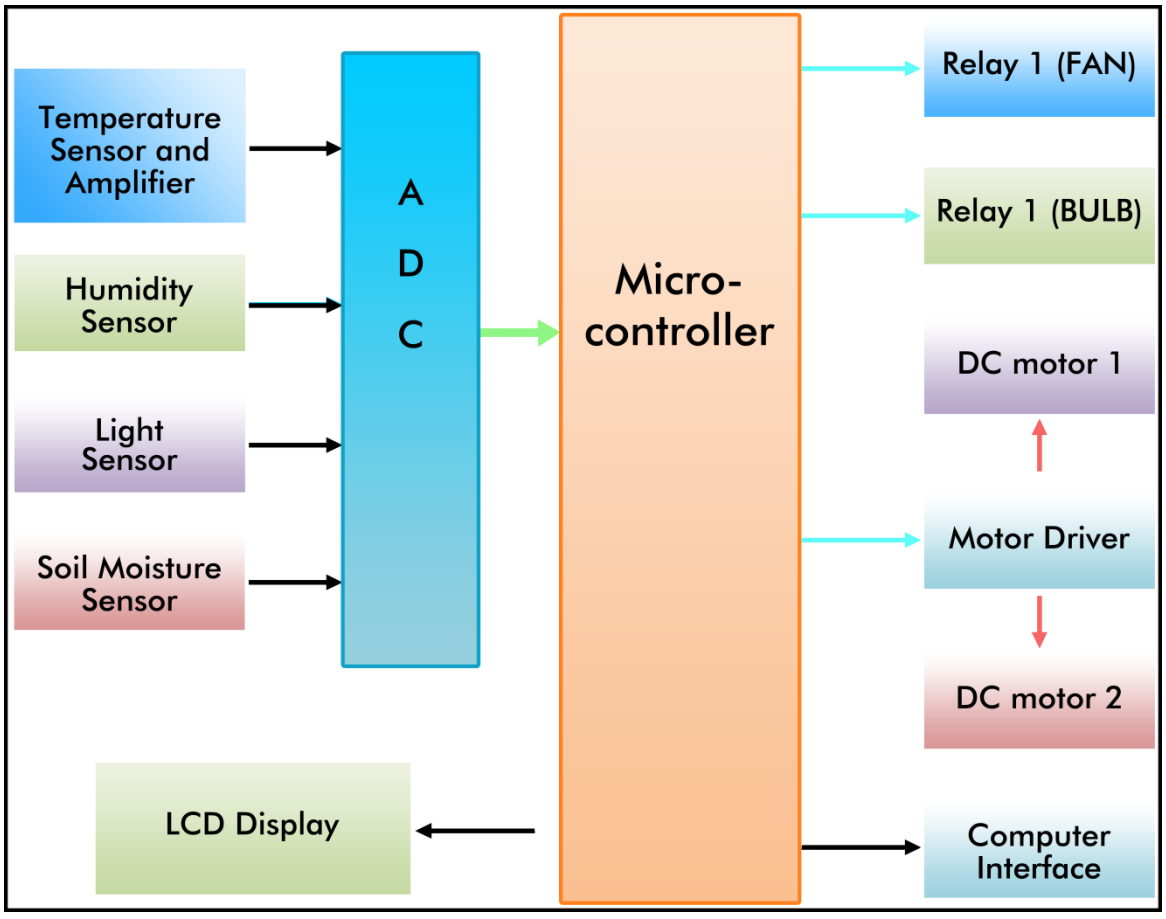


Figure 2.8: Weather Monitoring

2.4.2 Air Pollution Monitoring

The increasing number of vehicles on roads and rapid urban development has led to considerable increase in air pollution. Industrial development has further contributed to the deterioration of the air quality. Economic development has increased the number of vehicles on the roads, adding to air pollution problems with harmful smoke emission and other pollutants.

Wireless sensors and IoT systems used in air pollution monitoring systems can monitor the air quality emission of harmful gases, such as CO_2, NO2, CO, NO, and so on. Air pollutants mainly come from emission of gases and particles from vehicles, industries, and wood burning and from natural

sources that pollute the air, such as, dust storm, smoke from bushfire, volcanic activity, and forest fires.

Particle pollution is the term for a mixture of solid particles and liquid droplets found in the air. Primary particles are released directly into the air from sources (unpaved roads, construction sites, and vents) and from harmful gases (Sulphur-di-oxide (SO_2) and Nitrogen Oxides (NO_2)) that are released from power plants, automobiles, and industries. These particles contribute to air pollution. Real-time air quality monitoring system includes several distributed monitoring stations that communicate through wireless with a back-end server which interact with machine-to-machine. Each station is fitted with gaseous and meteorological sensors, as well as, data recording and wireless communication abilities. The back-end server gathers real time data from the stations and converts it into information, which is delivered to users through mobile applications and Web portals. Figure 2.9 displays air pollution emitting from the chimney of a factory.



Figure 2.9: Air Pollution

2.4.3 Noise Pollution Monitoring

Growth in urban development has led to increase in noise levels in metropolitan cities. Many environments, such as road works, airports, construction sites, factories, and so on producing loud noises, need effective noise pollution monitoring systems.

Noise pollution is a serious environmental crisis that affects people's health by increasing the risk of ischemic heart disease, hypertension, hearing loss, and sleep disorders (sleep apnoea and sleep insomnia), that also effect human productivity and behavior.

Wireless Sensor Network (WSN) offers inexpensive and flexible infrastructure to assist in the detection of noise pollution sources. An individual noise sensor is used to detect the Sound Pressure Level (SPL). It is a logarithmic measure of the effective pressure of a sound, measured in decibels (dB) from the surrounding environments. A number of noise monitoring stations are deployed in different places in the city that generate data on noise levels, which are collected on Web servers.

The collected noise level data is necessary for the preparation of noise maps which help policy makers in urban planning and drafting policies to control noise levels near important places in cities, such as, hospitals, schools, old age homes, parks, and so on. Smartphone users can continuously

measure noise levels in their surroundings, through Noise Tube applications installed in their phone and send it to a central server, where all information is aggregated and mapped to noise visualization maps that can foresee the intensity of noise pollution. Moreover, users can collectively participate in generating a map of noise pollution by sharing the data on their smartphone with the NoiseTube community.

Figure 2.10 shows the effects of Noise Pollution.

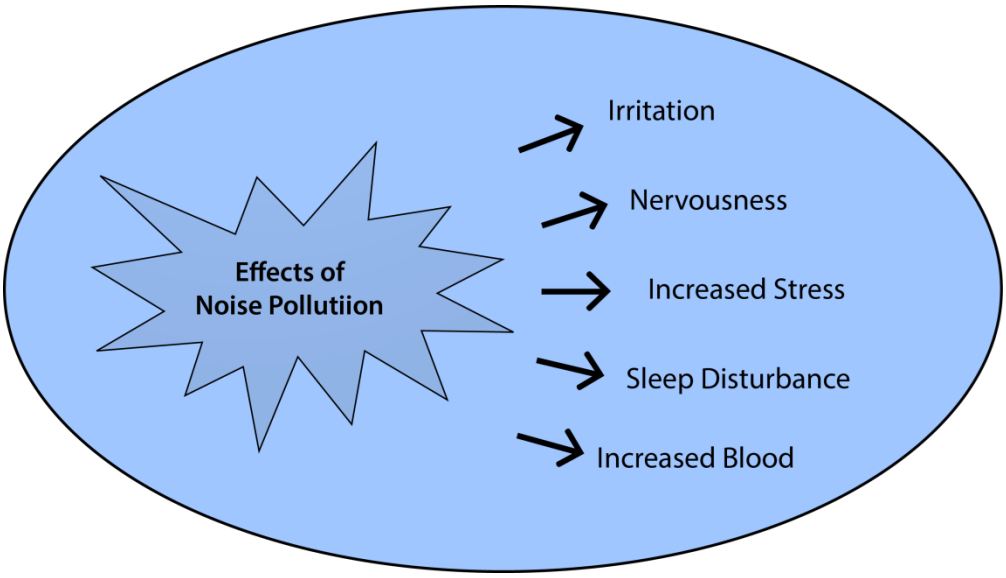


Figure 2.10: Depicts Effects of Noise Pollution

2.4.4 Forest Fire Detection

Forest fire is observed only when it has spread over a large area, making it arduous to control and stop it which is impossible at times. The result is destructive loss and permanent damage to the environment and atmosphere. Forest fires also have long-term devastating effects, such as impacts on global warming, local weather patterns, and extinction of rare species of plants or animal. The fire ignition may be caused through human actions, such as camp fire, arson, burning of debris, and so on and also from natural sources, such as, thunder and lightning strikes, volcanic eruptions, and fire sparks emitted from rock falls.

When the fire becomes uncontrollable, damage to the landscape will be inconsistent and the damage could last for a very long time depending on existing weather conditions and the terrain. Early detection of forest fires and monitoring of fire-prone areas can help in minimizing the damage, significantly shorten the reaction time, and also reduce the cost of fire fighting. IoT based forest fire detection systems use a number of sensors that are deployed at different locations in forest.

Sensors collect data on surrounding conditions including temperature, humidity, and intensity levels. Different types of detection sensors used are Infrared Rays, Video-cameras, Infrared spectrometers, Thermal imaging cameras, and Light Detection and Ranging Systems (LIDAR).

Figure 2.11 shows forest fire detection process.

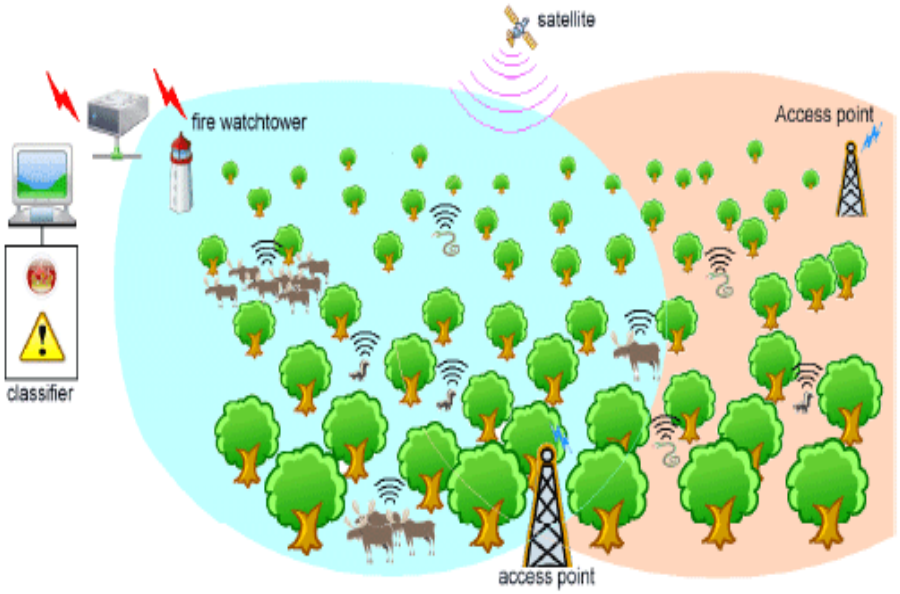


Figure 2.11: Forest Fire Detection

2.4.5 Flood Detection

Floods are caused due to continuous and heavy rainfall, which cause the river levels to rise and overflow rapidly and cause extensive damage to natural and human resources, and human life. An in-advance system of flood monitoring can avert risks and damages caused by floods. Flood monitoring system uses a number of sensors that are installed near river beds, which monitor the water level and flow rate. Data from a number of such sensor nodes is collected in Web server.

Monitoring applications send alerts when water level starts rising rapidly and over flows. Flood monitoring sensors enables detecting early flood signs, predicting floods, and monitoring and managing flooded areas. Sensors can be deployed along the bank of the rivers to measure increase in water levels and generate alerts by SMS or through Internet.

Sensors also monitor and observe weather conditions and rainfall, thus, enabling accurate prediction of floods and monitoring areas that are more prone to heavy flooding, such as, areas close to the river, coast, and so on. Measuring water absorption during rainfall and along the coastal areas will enable in determining areas where the flood risks are higher. This will enable in developing proper urban planning policy, by building viable rain-water pipe and sewage pipe system for high risk-prone areas.

During heavy rainfall, sensors can monitor areas vulnerable to flooding and can automate storm water drainage infrastructure to capture water and divert it to a different and safe path, so that, excessive water is not accumulated in given areas.
Sensors can provide prompt warnings and alerts for public evacuation in endangered areas in a suitable manner, in case the severity of floods is very high.

Figure 2.12 shows river flood detection.

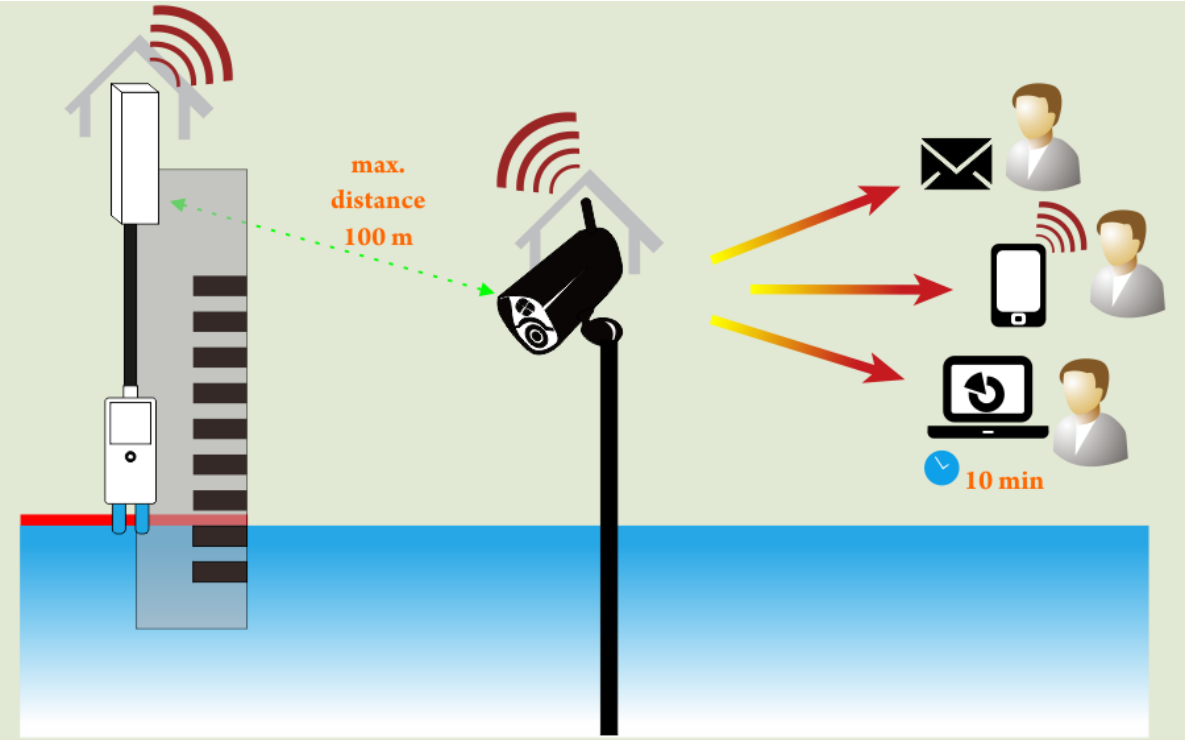


Figure 2.12: River Floods Detection

2.5 Energy

A smart energy system is a cost-effective, viable, and reliable energy system in which renewable energy production, infrastructures, and consumption are integrated and synchronized through active users, energy services, and enabling technologies. Smart energy system is an intelligent integration of decentralized renewable energy sources and power consumption that helps to form the future energy ecosystem.

2.5.1 Smart Grid

A smart grid is an interconnected network for delivering electricity from energy suppliers to consumers, which comprises energy operations, smart meters, smart appliances, renewable energy resources, and energy efficiency resources. Smart grid technology improves ability to detect fault faster and automatically rectify the fault through integrated network without the need for technicians. This enables consistent supply of electricity and decrease in vulnerabilities due to sudden failure of electrical grids.

Deployment of smart grid technology will improve efficiency of energy infrastructure; transmission and distribution infrastructure will manage bi-direction energy flows better; and reducing the voltage when possible on distribution lines. Smart grids technology uses integrated communications and sensors to enable real-time control, evaluating congestion, monitoring grid stability and health, energy theft prevention, and data exchange to enhance system reliability, asset utilization, efficiency, and safety of power delivery and use.

Smart meter is an automatic device that logs consumption of ‘electric energy in interims of an hour or less’ and transfers that data back to the energy supplier for tracking and billing. This will guarantee more detailed and precise energy bills.

Smart meters support two-way interaction between the meter and the energy provider. Smart meters can collect information for distant reporting. Figure 2.13 shows smart grid data communication network.

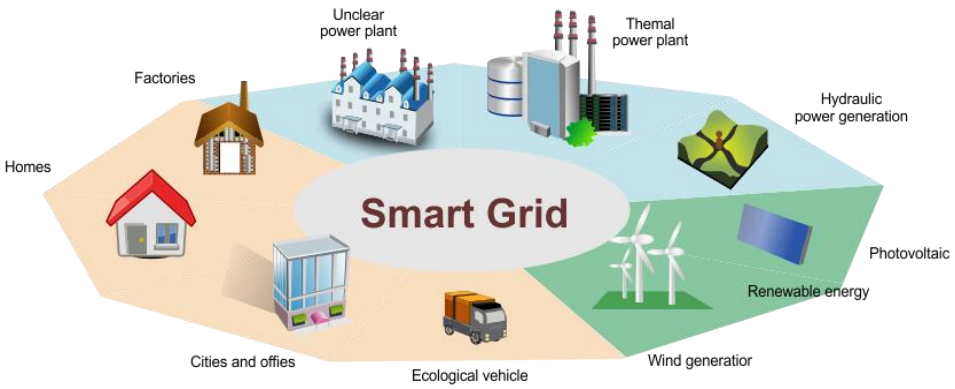


Figure 2.13: Smart Grid

2.5.2 Renewable Energy Systems

Energy from renewable source, such as, sun and wind are not always available. It is imperative that solar and wind energy supply is in proportion to the demand. This can be possible by the use of IoT based solutions. The smart grid monitors the generation, storage, distribution, and consumption of electricity. Wind turbines are the best substitute to solar energy.

To draw solar and wind power, integrating them into the grid enables more efficient use of solar and wind power and reducing fluctuations in energy supply, leading to optimized energy systems and reduced electricity costs. Variability in the output from renewable energy sources, such as, solar and wind, leads to voltage fluctuations that affect power quality.

Rapid expansion of renewable energy by developing existing technologies and investing in improvements in electricity system can enable better renewable energy output. The power lines, transformers, and control stations of old model energy grid are unreliable and not adequate to handle a significant increase in renewable energy. When renewable energy sources are integrated into the grid, transmission and distribution infrastructure will manage bi-direction energy flows better. IoT based systems integrated with transformers at the point of interconnection, measure the electrical variables and how much power is fed into the grid.

Figure 2.14 shows the Renewable Energy Systems.



Figure 2.14: Renewable Energy - Wind Turbines and Solar Panels

2.5.3 Prognostics

For energy systems to function efficiently, it is imperative that vital components that make up such energy systems must be in good working conditions. Components that form a part of energy systems, such as, smart grids, power plants, wind turbine, can perform their operations effectively, only when the state of their health is known periodically.

To prevent break-downs or failure in energy systems, due wear and tear of vital components, monitoring the energy systems and retrieving information in near real-time, enables forecasting break-downs or failures. Electrical grids collects, real-time information using specialized electrical sensors known as Phasor Measurement Units (PMU) that is installed in the sub-stations.

Wireless sensors deployed in energy systems periodically collect maintenance data in real-time, for monitoring the condition and the reasons for the break-down of energy systems so as to improve the consistency and availability of the energy systems. IoT based prognostic management systems that have been developed for energy systems can predict performance of machines and analyze the extent of deviation of a system from its normal operating conditions.

2.6 Retail

The IoT plays a vital role in the retail industry. Activities in the retail industry are now being substituted with IoT devices such as, sensors and RFID technology. RFID and sensors helps to overcome specific challenges in the retail industry. Technological developments, such as, inexpensive and smaller sensors, ubiquitous wireless networks, improved computing power, high-tech machine learning has made the IoT widely accepted in the retail industry and has revolutionized the retail sector.

The IoT helps retailers in offering superior product choice and improved customization. The IoT helps the retail sector to seal the digital-gap between consumer anticipations and retailers' potential to deliver. Emerging technologies has enabled retailers to transform customary ways of doing business and add new value in their businesses.

2.6.1 Inventory Management

Growing competition has necessitated the need effective Inventory management in retail sector. Over-stocking of products high storage costs, deterioration and obsolescence of perishable goods, loss due to sudden shift in demand for goods and low liquidity for businesses. Under-stocking can lead to loss of revenue due to lack of goods available to consumers and losing consumer loyalty.

IoT systems using RFID tags can help in inventory management and maintaining the right inventory levels. RFID tags provide a versatile way to track and control stocks. Products attached with can be monitored and managed in real-time, thus, facilitating in predicting the inventory level effectively. Therefore, under-stocking and over-stocking of goods can be avoided.

The RFID readers are attached to the retail store shelves and warehouses to enable easy tracking of data. IoT systems enable monitoring of inventory in real-time using the data collected by the RFID readers. In case inventory of any goods in a warehouse fall below the pre-defined limit, the RFID enabled system will place an automatic order and avoid an out-of-stock situation. Similarly, in case of over-stocking the system will generate alerts, which will enable taking decisions related to disposal of excess stocks. This will especially help in avoiding over stocking of perishable goods. As items are constantly monitored, Inventory shrinkages including misplacements, thefts of items can be prevented using RFID technology.

Figure 2.15 shows the flow of Inventory Management.

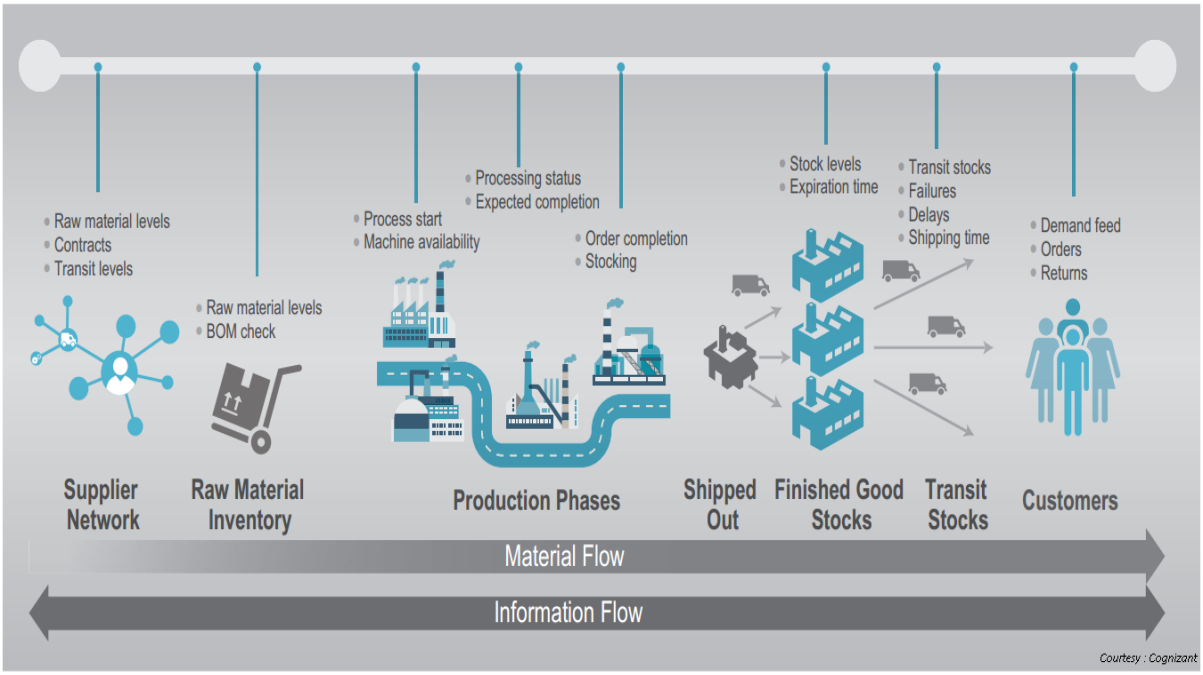


Figure 2.15: Inventory Management

2.6.2 Smart Payments

Smart payment solutions are contact-less payments, which uses RFID technologies, such as Near Field Communication (NFC) and Bluetooth to make secure payments. Debit cards, credit cards, smart cards, key fobs, smartphones and other mobile devices are various contactless payment systems.

The implanted chip and aerial enable users to wave their card, fob, or handheld device over a reader at the POS. The benefits of contact-less payment are delivering fast, secure transactions and enhance speed, security, and convenience. Users no longer have to struggle with cash and change or be concern about not having enough cash for purchasing any item and instead they can wave their wireless payment device in proximity to a reader and make payments. NFC is a form of mobile contactless payment. NFC comprises an initiator and a target; NFC initiates pairing of devices to establish a Bluetooth connection, while the actual data transfer takes place over Bluetooth. NFC is a one-step payment method that works anywhere the users want to make a purchase.

2.6.3 Smart Vending Machines

Smart vending machines connected to the Internet allow remote monitoring of inventory levels, promotions, and elastic pricing of products as well as contact-less payments via NFC. Sensors in a smart vending machine monitor its operations and send the data to the back-end which can be used for predictive maintenance. The advantages of smart vending machine are as follows:

- Allow users to transfer money from the bank and the fund is automatically stored in it. Users will be able to store funds in their account. When users purchase, the amount of the product that they will be buying is automatically deducted from the fund. This is advantageous for users especially when they do not have cash in the pocket.

- Give users the history of their past purchase. It enable users to know when they have been purchasing, how many times they have purchased the same product, and history of their other purchases - which will all appear on the screen. With this, users will know what sort of transaction they had in the past and it helps users to remember the number of times they had used the vending machine.

- Automatically give users an altered offer according to what they want. Since, the machine is tracking the purchases that users made in their last few transactions, it has the ability to save the history of food bought by the users. The next time users buy food they would be given a list of choices from the ones that they had been buying in the past.

- Smart vending machines can communicate with other vending machines in their vicinity and share their inventory levels, so that customers can be routed to the nearest machine in case a product goes out of stock in a machine. For perishable items, the smart vending machines can reduce the price as the expiry date nears. New products can be recommended to the users based on the purchase history and preferences.

2.7 Logistics

Supply chain and logistics enabling technologies have played an important role to network people, information, and products over conventional supply chain boundaries comprising management of various industrial, logistics, and retailing functions, such as, warehousing, manufacturing, and distribution of goods.

Therefore, optimization of models integrating with Web-based systems, as well as, smart materials characterized by actuators and sensors, tagging represented by GPS and RFID, and decentralised processing amongst trading associates can build innovative manufacturing, logistics, and distribution processes that offer a dynamically structured supply chain network.

2.7.1 Route Generation and Scheduling

IoT based systems for route generation and scheduling enables optimization of multiple route generation, dynamic online route planning, and optimization of the routes. Data from various IoT based technology provide services, such as, dynamic vehicle muting, advanced route guidance, and foreseeing customer demands for pickup and delivery issues.

As the transportation network grows in size and complexity, the number of routes has increased enormously, thus, requiring the need for proper route management and scheduling. Route generation and scheduling generate end-to-end routes using permutation of route patterns, transportation modes, and feasible schedules based on the availability of vehicles. IoT based systems

enables optimizing transportation schedules, planning fixed routes, strategic logistics planning, route optimization and fast response to the route generation queries.

2.7.2 Fleet Tracking

Vehicle tracking systems are used by fleet operators for fleet management functions, such as, tracking of fleet, scheduling the best routes, dispatching of goods, information of fleet during transit, and for monitoring security of fleet. Modern vehicle tracking systems commonly use GPS technology for locating the vehicle. Vehicle tracking systems for fleets enable real-time monitoring of vehicles, reduce fuel consumption, cut idling and emissions, and eliminate inefficiencies in fleet. Vehicle information can be viewed in real-time using the Internet or IoT.

Vehicle tracking system enables monitoring vehicles routes and at the same time charts the most efficient routes. Cloud-based GPS tracking solution can identify fleet inefficiencies and cut costs significantly. Fleet tracking system provides a wide range diagnostics and maintenance reports on vehicle faults, thus, enabling fleet to stay safe on the road.

Vehicle tracking system also tracks drivers' movement and their driving behavior, thus, facilitating monitor and controlling drivers from over speeding or risk driving. In case of breakdown of vehicle during transit, GPS technology identifies the next fleet nearest to the break down vehicle and diverts the fleet to the destination. This prevents any delay in delivering goods to the concerned parties, thereby building goodwill and strong customer relationship.

2.7.3 Shipment Monitoring

Shipment monitoring technology enables manufacturers and shippers to track shipments, monitor inventory during shipment, and prevent theft of shipments. Technology such as RFID, sensors facilitates information of shipment, location tracking, status of loads, and exigencies. IoT devices enables communication and tracking information in real-time. These systems are cloud-based, and can be tapped from any device with an Internet connection.

The system allows tracking cargo, investigating alerts generated sensors, and measures to tighten cargo security. Shippers can receive information about shipment location, idling time, and cargo loading status. Sensors detect light and temperature, pressure, humidity, and so on to monitor the conditions inside the containers and send the data to the cloud, where it can be analyzed to detect food spoilage.

Sensors identify if packages are tampered with and send E-mail alerts to concerned authorities to take remedial actions promptly and enhance security. Mobile apps allow shippers to input cargo-related data, from where data is transmitted to a central database, to be viewed by authorized parties. Cargo container fitted with GPS and RFID effectively tracks units or pallets.

2.7.4 Remote Vehicle Diagnostics

Remote vehicle diagnostic systems can detect faults in the vehicles or warn of impending faults. The system monitors the condition of the vehicle, identifies the causes, and offer real time information of vehicle parameters to assess its performance. The solution monitors the health of vehicles and enables technicians to determine the main cause of the problem.

Sensor and microprocessor enable remote diagnostics in a vehicle by providing in-vehicle data on vehicle operation, such as, speed, engine Revolutions per Minute (RPM), coolant temperature, fault code number, and status of vehicles' ancillary systems.

IoT based vehicle diagnostic systems can send the vehicle data to centralized servers where it is analyzed to generate alerts and suggest remedial actions. Technicians can view diagnostic information in real-time from the vehicle and chart a maintenance plan with detailed information and the time needed for repairs, perform remote diagnostics and learn automatically from errors. Real time information on vehicle fault codes facilitates swift and precise diagnostics, this enabling to carry out the necessary repairs efficiently and without any delay.

2.8 Agriculture

The IoT is changing the agriculture industry and enabling farmers to cope with the huge challenges they face. Innovative IoT applications are focusing on these challenges and increasing the quantity, quality, cost efficiency, and maintainability of agricultural production.

IoT facilitates controlling and monitoring the agricultural industry with sensors that can detect, crop growth, soil moisture, cattle feed levels, monitor irrigation equipment and harvesters, and analyze operational data together with third party information and data, such as, weather services, to offer new visions and enhance decision making.

2.8.1 Smart Irrigation

Smart irrigation systems use IoT devices with soil moisture sensors, to determine the amount of moisture in the soil and release the flow of water through the irrigation pipes when the moisture levels goes below a pre-defined limit. Low-cost wireless sensors are sown into fields to measure moisture at various depths below the ground. This information or data will be emitted through wireless transmission to a central hub, which directs water to the crops that require it and that best penetrate the ground with least waste.

Smart irrigation systems also gather moisture level measurements on a server or in the cloud, where the gathered data can be analyzed to plan watering schedules. Information or data on soil moisture, temperature, humidity, soil dryness, can be retrieved through Internet and wireless network. Real-time monitoring of field information enables to assess the crop water demand, activate the field irrigation automation, and ensure accurate water supply to crops and enhance yields.

2.8.2 Green House Control

Greenhouse monitoring and controlling is used to measure various factors, such as, temperature, humidity, light, and soil moisture. Microcontrollers' monitors and stores the values of temperature, humidity, soil moisture, and sunlight of the natural surroundings that are periodically altered and controlled in order to optimize them to achieve maximum.

The microcontroller communicates with the various sensors in real time in order to control the light, aeration, and sewage system effectively inside a greenhouse by triggering the fan cooler and lights according to the condition of the crops. IoT systems play a significant role in green house control and enable in enhancing productivity. The data collected from various sensors is stored on centralized severs or in the cloud, where analysis is performed to optimize the control strategies and also, correlate the productivity with different control strategies.

2.9 Industry and Manufacturing

The IoT will revolutionize ways in which industries and manufacturer do business. Each industry will profit from investing in the IoT. Data-collecting sensors in office buildings, factory machinery, and warehouse shelves will enable monitoring and controlling of business efficiently and will reduce cost.

The IoT will transform industries by opening up a new era of economic growth and competition. The IoT will have a positive impact on the operational efficiency, productivity, and operations of industries. Industries can also benefit significantly by perceiving it as a tool for development in unanticipated opportunities. The IoT will increase revenues by increasing production and constructing new hybrid business models, develop intelligent technologies to power innovation, and bring transformation in workforce.

2.9.1 Machine Diagnosis and Prognosis

IoT plays a crucial role in both projection and diagnosis of industrial machines. Machine prognosis refers to anticipating the performance of a machine by scrutinizing the data on the present operating conditions and inconsistency or variation from the normal operating conditions. Machine diagnosis refers to determining the reason for a machine's defect.

Machine diagnosis and prognosis is becoming critically important across industrial and commercial applications. The operational demands and high procurement costs of machineries for industries require a high degree of uptime and high reliability. Unexpected failures are costly to correct and such failures will be catastrophic. Industrial machines have numerous components that must function perfectly, for the machine to perform its operations accurately.

Sensors in machines can monitor the operating conditions, such as, temperature and vibration levels. IoT based systems integrated with cloud-based storage and analytics back-ends can help in storage, collection, and analysis of machine sensor data. This data enables in effectively monitoring machines and take remedial actions in case of faults or failure or breakdown effecting the operational functions of the machine.

2.9.2 Indoor Air Quality Monitoring

Indoor air quality in Industries can profoundly affect the health of workers and productivity of work force. Harmful and toxic gases, such as, CO2, CO, NO, NO2, and so on, can cause serious health issues. Monitoring indoor air quality in factories is imperative for health and safety of the workers. Sensors are deployed in different places within the factory for monitoring the air quality.

Sensors can identify the hazardous zones for poor ventilation and enable remedial action to ensure proper airing to dilute contaminants in the air. Sensors and wireless networks also monitors the environmental factors, such as, humidity, temperature, vaporizer, particulate matter, and gaseous pollutants, which enable effective control of the indoor air quality.

2.10 Health and Lifestyle

The importance of healthy lifestyle has driven innovation to play an important role in self-care. Enabling technologies enables periodical capturing of information and data related to an individual's health, lifestyle, and other vital parameters and sharing it with health practitioners.

The IoT enables the real time capture of information, assists collection and analysis of the information, thus, shifting the point of care from the hospital to the patient's home. IoT will help to resolve the industry's existing challenges and radically improve health results at lower cost. IoT is an innovation that changes the ability to exchange and make use of information related to health of individuals. IoT will change the way in which healthcare will be provided, driving better results, increasing efficacy, and making healthcare inexpensive. The ability of patients to take responsibility of their health and the promotion of preventive measures through IoT will shape the future of healthcare.

2.10.1 Health and Fitness Monitoring

Wearable IoT devices enable uninterrupted health and fitness monitoring. These wearable devices can be in different forms, such as, belts, watches, goggles, and wrist-bands. The wearable devices have sensors implanted in them, which measures various health parameters periodically and data from these devices are constantly sent to a central server for analysis and archiving. Healthcare providers can scrutinize the collected healthcare data to assess any health conditions or irregularities.

The influence of home-use medical devices and mobile healthcare applications has been increasingly felt in the healthcare market. Apart from their use in monitoring health, they have also been used for prevention of disease and promotion of fitness. Some of the applications of IoT are:

- IoT devices monitor patient's physiological conditions, such as, blood glucose levels and weight for diabetics, blood pressure for hypertensive patients, and so on. The intermittent data sent in by these devices can be used to perform various analytics.

- IoT transfer the tasks of regular monitoring and field management to remote monitoring personnel, thus saving cost. Moreover, remote monitoring has effectively demonstrated results, such as, increased staff productivity and decreased member move-out rates.

- IoT, as wearable technology, has made health supervision expedient. Such incessant availability of data enables the user to analyze and compare many parameters.

Common use body sensors has, Pulse Oximeter Oxygen Saturation (SpO_2), body temperature, Electro-Cardiogram (ECG), heart rate, movement with accelerometers, blood pressure, and Electro-Encephalogram (EEG).

Smart shoes monitor the walking or running and jumping speeds, with the aid of implanted sensors, and are synchronized with smartphones to visualize the data. Smart wristbands can track the daily exercise and calories burnt. Figure 2.16 shows patients with smart wearable devices for health monitoring.



Figure 2.16: Depicts Patients with Smart Wearable Devices for Health Monitoring

2.11 Summary

- Growth of Wi-Fi's role in home automation has mainly increased due to the integrated nature of installed electronics, where electronic devices (TVs and AV receivers, mobile devices and so on) have started to play a vital role in the home IP network.
- Present homes have different types of modern appliances, such as, air-conditioner, music systems, washer/dryers, refrigerators, TVs, and so on. Handling and governing these appliances can be arduous, as each appliance is designed with unique controls or remotes.
- Smart parking is driven by IoT systems which perceive the number of vacant parking spaces and send the information through the Internet to smart parking application back-ends.
- IoT systems for fire detection, gas, and water leakage detection can help in generating alerts and minimizing their effects on the vital infrastructure.
- IoT based weather monitoring systems can collect data from a number of sensor attached, such as, temperature, humidity, pressure, and so on. It can also send the data to cloud-based applications and storage back-ends.
- IoT systems using Radio Frequency Identification (RFID) tags can help in inventory management and maintaining the right inventory levels.
- Wearable IoT devices that allow non-invasive and continuous monitoring of physiological parameters that can help in continuous health and fitness monitoring. These wearable devices can be in various forms such as belts and wrist-bands.

2.12 Check Your Progress

1. Which of the following technology plays a major role in home automation?
   a. Growth of wired tech
   b. Growth of IR
   c. Growth of Wi-Fi
   d. Growth of LAN

2. Which among the following device assist in home intrusion detection?
   a. Security camera
   b. Wi-Fi modem
   c. Smoke detector
   d. Smart TV

3. Which sensor assist in IoT based smart irrigation?
   a. RFID
   b. Soil moisture sensor
   c. Smoke detector
   d. Smart clip

4. Which among the following option is widely used in IoT based inventory management?
   a. Wi-Fi Hotspot
   b. IR Tags
   c. Bluetooth Tags
   d. RFID Tags

5. Which among the following is a wearable device?
   a. Smartphone
   b. Google Glass
   c. Tablet
   d. Laptop

2.12.1 Answers

1	**C**
2	A
3	B
4	D
5	B

SESSION 3 - DESIGN AND STRATEGY FOR CONNECTED DEVICES

Session Objective:
This session explains the Design and Strategy of connected devices and explains in depth Radio Frequency Identification (RFID). Also, this session will discuss the importance of networking in IoT and an overview of network protocols. The session further will focus on Logical design of IoT and various computing devices.

At the end of this session, students will be able to:
- Explain Radio Frequency Identification (RFID)
- Explain the importance of Networking in IoT
- Explain Network Protocols for IoT
- Explain the Logical Design of IoT
- Describe Computing Devices

3.1 Introduction

The role of IoT is very significant in the present Information and Technology world, considering its vast application across various domains such as businesses, logistics, health, transportation, homes, manufacturing, and so on. Heterogeneity of domains has resulted in each domain requiring different standards and IoT systems with different performance expectations. The design and architecture of the IoT has been influenced by the diversity of domains, ensuing in varied IoT components with different functionalities that comply with specific domain requirement. IoT devices have distinct identities and can perform remote sensing, actuating and monitoring functions. IoT devices mutually exchange (collect and send) data with other connected devices and applications. IoT device has several wired and wireless interfaces for connection to other devices. These interfaces are interface for sensors, Internet connectivity, memory and storage, and audio/video. Data from these interfaces are collected by IoT, which are processed either locally or in the cloud-based application back-ends. IoT devices are of different types such as wearables, sensors, lightings, surveillance system, automobiles, industrial machines, and so on. The data generated by the IoT devices are processed by the data analytics system, which facilitates in directing further activities remotely or locally. IoT devices can be connected to actuators that enable them to interact with other physical entities, including non-IoT device and systems in the vicinity of the devices.
IoT system solution includes a Gateway board, which controls the entire system and several sensor nodes. The data from the different sensor nodes can be seen on the Web application over cloud which can be viewed by the user with help of smartphone anywhere and at any time.
Figure 3.1 shows the block diagram of the IoT system.

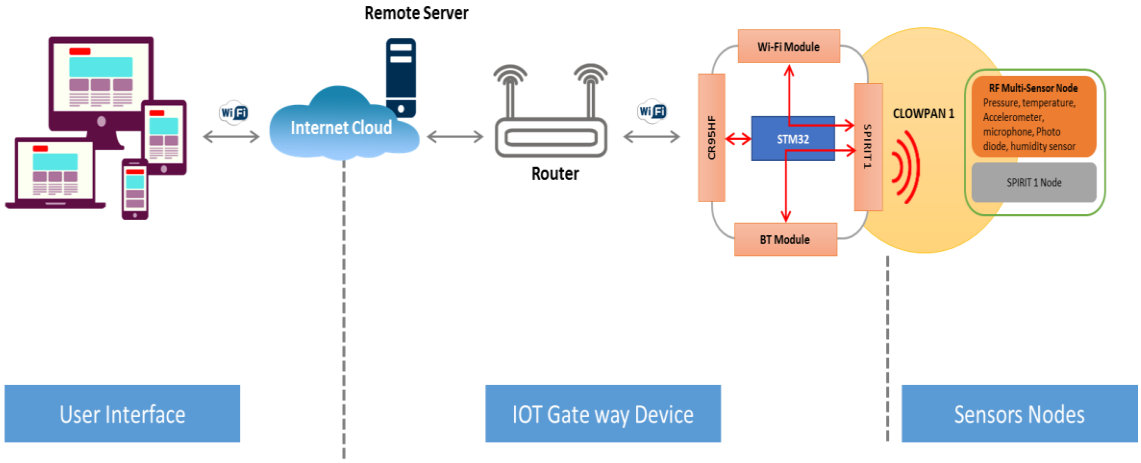


Figure 3.1: IoT System Block Diagram

3.2 Radio Frequency Identification (RFID)

Radio frequency Identification are IoT devices, which uses radio frequency or waves or electromagnetic fields, to peruse and encapsulate information stored on a tag, which are either embedded or attached to objects, or things, or people. RFID tags comprise an aerial that helps in transmitting to and receiving signals or radio waves from the RFID reader. The RFID tags also have a microprocessor or microchips, where information about the objects and things are stored and can be processed for identification and information by the RFID reader. The 'interrogator or reader' direct a signal to the tag and read (infer) its response.

Figure 3.2 shows RFID Tag.

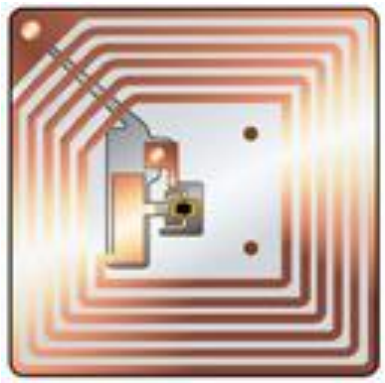


Figure 3.2: RFID Tag

RFID tags can be classified as passive and active tags. Active RFID tag is equipped with a battery of its own that enables them to have long read ranges as well as very large memory. Active RFID tags when powered by battery automatically broadcast their signal. A passive tag is cheaper compared to an active tag. Passive tags, unlike active tags do not have power source of their own and need to use the radio signals or radio waves diffused by the RFID reader. The passive tags transmits signal back to the reader, once the RFID reader emits radio waves for the reader to process information from these tags. However, the RFID reader can read only those passive tags that are in proximity to the RFID reader. The tags comprise electronically stored data. RFID tags can be tags where the data cannot be modified (read-only), comprising a manufacturer allotted 'code' which is used as a key into a databank, or separate data can be written into each object by the system user (read/write).

The tag data is kept in a Non-Volatile Memory (NVM). NVM is type of memory that is capable of holding data saved in it without the data being lost and does not require the memory contents or data to be refreshed periodically.
Figure 3.3 shows an Active Tag and Passive Tag.

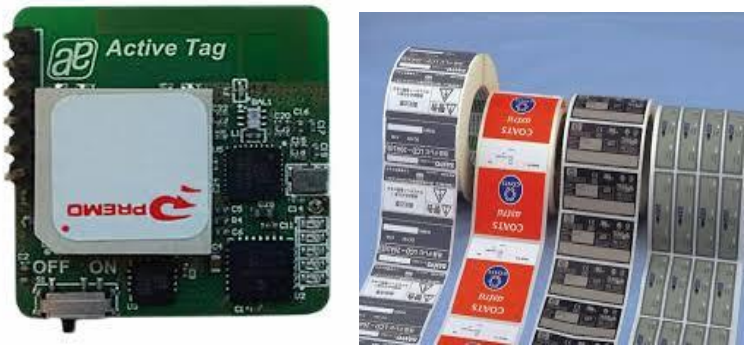


Figure 3.3: An Active Tag and Passive Tag

RFID reader diffuses radio waves or electromagnetic waves to the RFID tags in order to read or process the RFID tags. The RFID tags respond back once the tags accept or receive the radio signals from the RFID reader. Each tag has a unique identification number which enables RFID reader to distinguish amongst various RFID tags-both active and passive. RFID tags can read several hundred tags simultaneously.
Figure 3.4 shows RFID Technology.

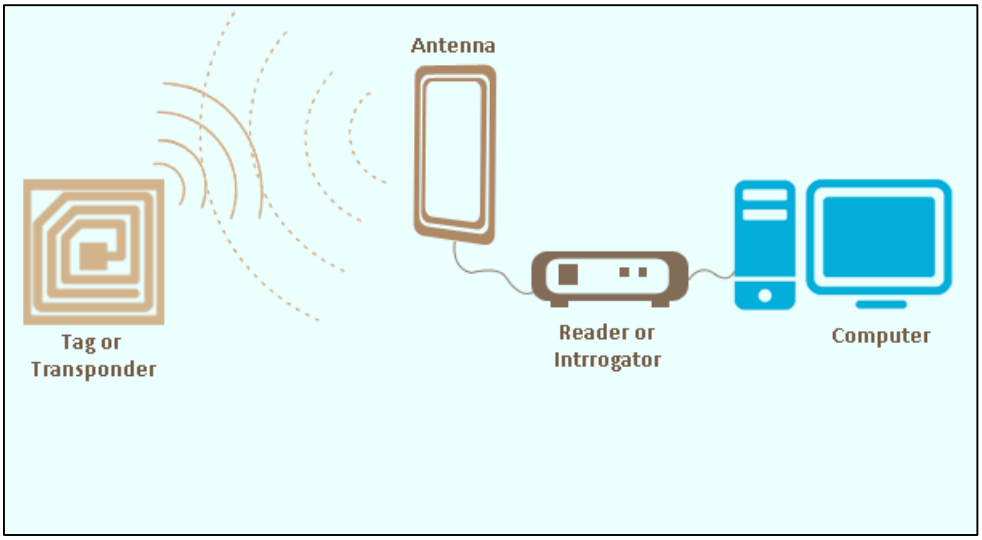


Figure 3.4: RFID Technology

RFIDs are easy to incorporate or install in other objects. The tag can be read even if the tags are concealed or covered by the things or items. Tags can be read at few hundred feet distances or more and have sensors implanted in them that can use electricity for power. Since, RFID tags can be affixed to currencies, apparel, and belongings or it can be imbedded in living beings. Extracting confidential or personal information without approval has raised serious privacy and security issues and concerns.

Figure 3.5 shows various applications of RFID.

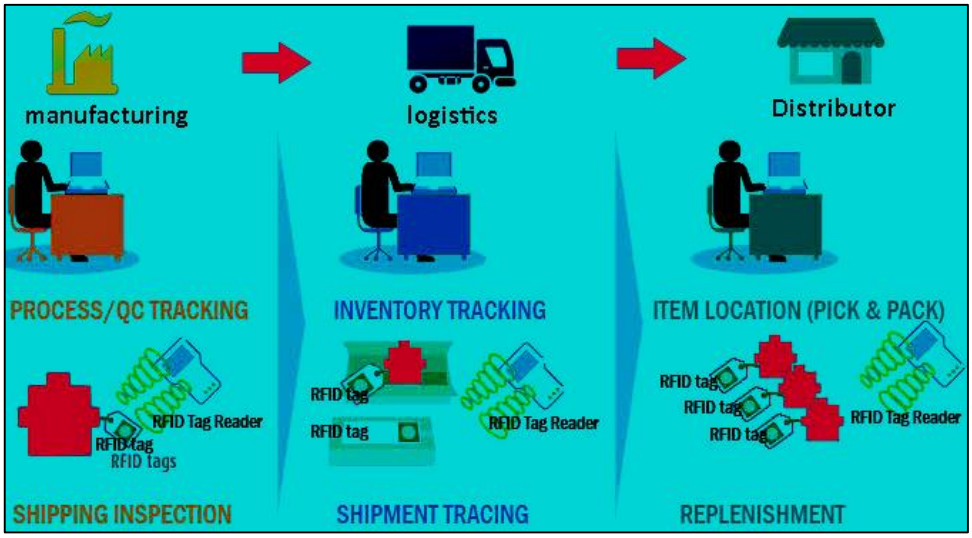


Figure 3.5: RFID Applications

3.3 Importance of Networking in IoT

IoT has sensors and gadgets that collect huge volumes of data and such voluminous data need to be evaluated for its implications on the network and networking industry.

A dynamic and secure networking is imperative for IoT. Areas that will influence the IoT network are as follows:

- **Data Analytics**: The inflow of sensors will generate huge volumes of data that needs to be processed in a cloud-based data center, as moving data effectively the cloud is very crucial. The volume of data produced by the IoT will have an impact on the networking systems.
- **Agile Networking**: Quick and agile networks are vital to enable the real-time analysis of sensor data.
- **Security**: Being constantly connected to people and systems through devices such as smartphones, tablets, and computers will endanger the users in many ways. It is imperative that the networking industry guarantee safety and guard against unavoidable threats within these networks.

3.4 Architecture of a Network for Connected Devices

Several networking features have to be considered when selecting wireless networks which are as follows:

- **Latency**: It is defined 'as the time taken for a packet of data to move from the sensor node over the network to the gateway node and back from the gateway node to the sensor node is known as latency.' In general, latency refers to the swiftness of the network; the quicker the network, the lesser the latency.
- **Throughput**: It is defined as 'the volume of data that can pass through the network, per second is called throughput'.
- **Fault resiliency**: It is defined as 'the degree to which a wireless network, if interrupted, will retrieve or reconfigure, and transmit a packet of data to its destination'.
- **Scalability**: It is defined as 'the number of nodes that can be encompassed in a particular network is defined as scalability'.
- **Hops**: It is defined as 'the transmission of a packet of data from one node to another is called hops'. The 'number of hops' refers to the number of nodes through which the data moves.

- **Range**: It is defined as 'the range of the network is the total distance a whole network can cover. The range of a node is the highest distance of one hop, from one node to the other'.

3.4.1 Types of Architectures

The networking standards in IoT can be classified into three elementary network topologies; Point-to-Point, Star, and Mesh.

- **Point-to-Point Network**: A point-to-point network creates a direct link between two network nodes or connects two nodes directly together. Transmission can take place only between these two nodes or devices and link between the two nodes is permanent. For example, Bluetooth connects a mobile and an ear piece.
- **Star Network**: A star network comprises one main hub to which all other sensor nodes in the network are connected. The hub provides a common connection point for all nodes and data transmission between nodes take place through the central hub. The central hub can communicate with hubs of different network by transmitting to and receiving from the central hub. For example, WiFi or Ethernet.
- **Mesh Network**: Mesh network nodes are installed in a way that each node is connected to every other node in the network. Each individual node is in communication range of at least one other sensor node. The nodes transmit data for the network and the data pass through multiple sensor nodes to reach the gateway node. This networking topology is utilized for applications needing a long range/broad area coverage.

  Figure 3.6 shows the three network topologies for IoT.

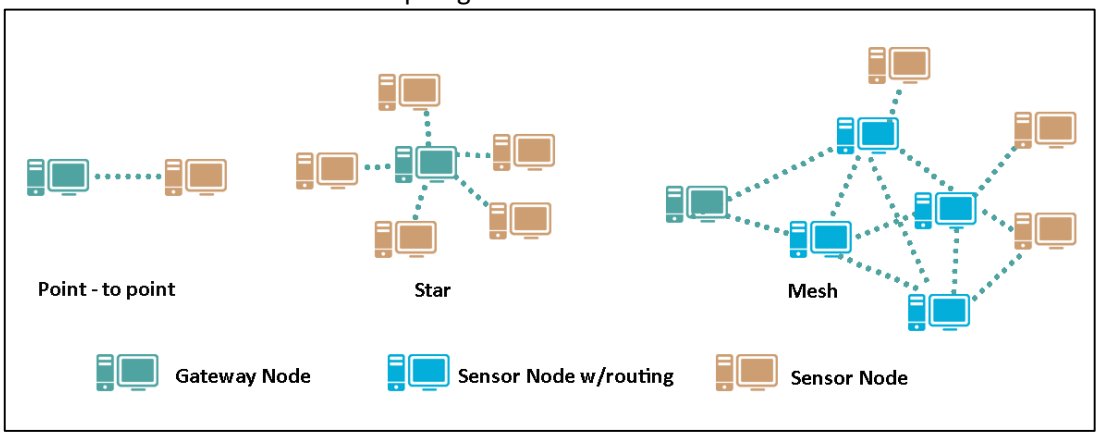


Figure 3.6: Three Networking Topologies for IoT

3.5 An Overview of Various Network Protocols

Network protocols are set of rules, procedures, and formats for sending and receiving data successfully between network devices. Network protocols comprise mechanisms for devices to identify and make connections with each other, send and receive messages in the form of packets, and manage data in a timely and secure manner.

3.5.1 Types of Network Protocols

The different types of network protocols are as follows:

- **802.3 Ethernet**: Ethernet is a part of computer networking technologies and act as a physical and data link layer for wireless Local Area Networks (LANs) and Metropolitan Area Networks (MANs). Ethernet defines the ways various networked devices can format data for transmission to other network devices. Data travels over Ethernet inside protocol units called frames. Ethernet uses a star topology and is one of the most widely implemented LAN standards.
- **802.11 WiFi**: WiFi is a local area wireless computer networking technology that enables computers, smartphones, or other devices to connect to the Internet and communicate with one another wirelessly within a specific area. WiFi uses radio waves to provide wireless high-speed Internet and network connections.
- **2G/3G/4G - Mobile Communication**: 2G/3G/4G are second/third/fourth generation mobile technology. The primary benefits of these mobile technologies are mobile conversations are digitally encoded, enabling the transmission of data to only the intended receiver. 2G technologies provide services such as text messages, Multimedia Messages (MMS), and picture messages. 3G can be used in wireless voice telephony, video calls, fixed wireless Internet access, mobile Internet access, and mobile TV. 4G system offer capabilities such as advanced mobile Web access, 3D television, high-definition mobile TV, video conferencing, IP telephony, cloud computing, and gaming services.
- **Network/Internet Layer**: The network layer is a part of the Open Systems Interconnection Model (OSI Model) and the layer (third layer) that offers data transmitting paths for network communication.
  It chooses and manages the best rational (logical) path for data transmission between nodes. This layer comprises hardware devices (physical medium) such as firewalls, switches, bridges, and routers.
- **IPv4:** Internet Protocol version 4 (IPv4) is a wireless protocol of standards-based networking techniques in the Internet that makes it possible to connect devices to the Web. IPv4 is used on packet-switched networks. However, the limitation of IPv4 is its addresses are becoming scarce and IPv4 does not assure delivery, nor does it guarantee accurate sequencing or prevention of identical delivery.
- I**Pv6**: Internet Protocol version 6 (IPv6) the descendant of IPv4, which enables address allocation on the basis of ranking or hierarchy and directs traffic across the Internet. Unlike IPv4, IPv6 solves the problem of address depletion and has sufficient addresses to keep the Internet operational. IPv4 routes the most Internet traffics in spite IPv6's advantages.
- **Transport Layer**: The transport layer is the layer of the OSI model that ensures transfer of data between end systems. It offers rational path for communication between application processes operating on different hosts and decides if data transmission should be on parallel path or single path. The transport layer ensures receiving of whole message and in order, ensuring both error and flow control at the source to destination level. This layer breaks the data into small units in the form of packets so that they are handled more efficiently by the network layer.
- **TCP**: Transmission Control Protocol (TCP) is a protocol designed to transfer data across the Internet by breaking the data into smaller packets. TCP is a layer used by applications that require data reliability and guaranteed delivery across supporting networks and the Internet. TCP establishes two-way virtual connection between two endpoints (source and destination node) and retains the link until the communication is active.
- **UDP**: User Datagram Protocol (UDP) is a protocol similar to TCP, in which data is transmitted across the network, to the receiver, without prior arrangement between the sender and receiver (connection-less). It is used mainly for broadcasting messages across a network.

- **HTTP**: Hyper Text Transfer Protocol (HTTP) is a protocol used by the World Wide Web (WWW). HTTP sets frameworks for Web servers and browsers to take the required actions and execute the necessary commands sent through the network. HTTP is a stateless protocol, since, each command is executed independently and without the prior knowledge of other commands.
- **CoAP**: Constrained Application Protocol (CoAP) is a protocol meant to be used in simple electronic devices and limited resource Internet devices such as remotely controlled WSN nodes, small switches, and so on, thus, permitting interactive communications across the Internet. CoAP is an application layer protocol and can operate on devices that support UDP.
- **WebSocket**: WebSocket is a self-supporting TCP-based protocol offering transmission of two signals simultaneously in the opposite directions across a single TCP connection. The WebSocket protocol makes communication between a browser and a Website possible by enabling the real-time data transmission from and to the server.
  The WebSocket protocol is supported by browser such as Firefox, Opera, Safari, Internet Explorer, and Google Chrome. WebSocket also needs Web applications on the server to back it.
- **AMQP**: The Advanced Message Queuing Protocol (AMQP) is an open IP for message-oriented middleware and enables messaging providers and users to interoperate seamlessly. The AMQP perform services, such as, routing, queuing, message orientation, security, and reliability and defines how these services take place between the messaging provider and user. Messages from different middleware products, written in different languages and using different platforms, can be transmitted to each other through AMQP.
  Figure 3.7 shows the Network Protocols.

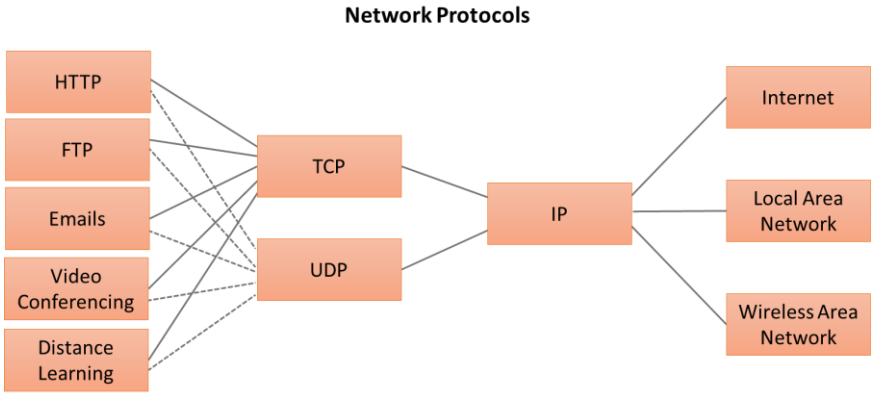


Figure 3.7: Network Protocols

3.6 Logical Designs of IoT

'Logical design of IoT is a tangible concept in which data is arranged by programmers in a sequence of logical relationships or entities'.

The steps involved in the design methodology of the IoT are:

- **Purpose and Requirement Specifications**: The purpose, behavior, and the requirements of the system are summarized. Requirements comprises data collection, analysis, privacy, and security. Systems management requirements and user interface requirements are also needed.
- **Process Specification**: It defines the use of the IoT systems based on the requirements and purpose specifications of the IoT.
- **Domain Model Specification**: The domain model describes the main objects, concepts, and entities in the domain of IoT system.
- **Information Model Specification**: Defines the attributes of Virtual entities, relationship, and structure of all models in the IoT.

- **Service Specifications**: Defines the service - types, inputs/output, endpoints, schedules, pre-conditions, and effects.
- **IoT Level Specification**: Define the IoT level for the system.
- **Functional View Specification**: It defines the functions of the IoT systems.
- **Operational View Specifications**: This specification defines system development and operations, such as, service hosting option, storage option, device option, and application hosting option.
- **Device and Component Integration**: Defines the assimilations of devices and components.
- **Application Development**: Develops the IoT Application itself.

3.6.1 Functional Blocks

An IoT system comprise number of functional blocks that provide the system abilities for communications, identifications, actuating, sensing, and management. The functional blocks are:

- **Device**: An IoT system comprises devices that provide control, sensing, actuation, and monitoring functions.
- **Communication**: The communication block manages the communication for the IoT system.
- **Services**: An IoT system uses various types of IoT services for data publishing, Device - monitoring, control, and discovery.
- **Management**: This functional block offers numerous functions to the IoT system.
- **Security**: This functional block safeguards the IoT system by offering and providing functions such as authorization, message and content integrity, authentication, and data security.
- **Application**: IoT applications can provide an interface that users can use to control and monitor various aspects of the IoT system. Applications allow the users to view the system status and view or analyze the processed data.

3.6.2 Communication Models

There are four Communication models for IoT Functional Components and they are as follows:

- **Push-Pull Pattern**

  The Push-Pull pattern is a one-way communication in which the server sends data to a pre-defined client who receives the data. The address of the client is known beforehand by the server and the client is also periodically expecting data from the server. In this kind of communication model the communication partners do not change frequently. Figure 3.8 shows the Push-Pull Communication model.

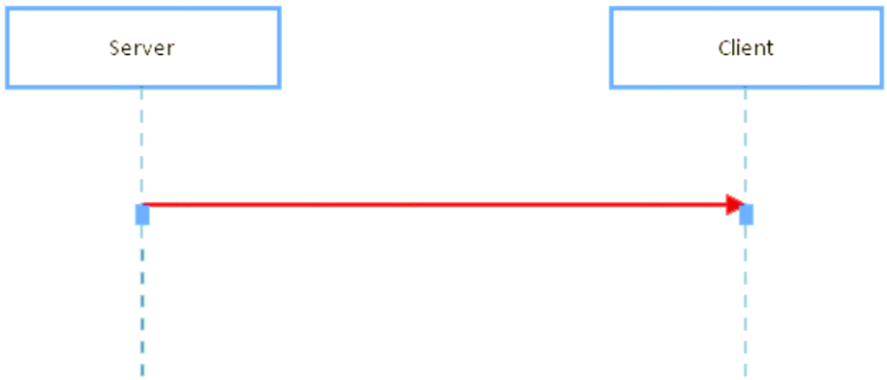


Figure 3.8: Push- Pull Communication Model

Request-Response Model

The Request-Response model is a model in which a client sends a request to a server and the server will respond to the request. The client will have to wait for the response from the server, since, the server will take time to respond to the client's request. In the meanwhile if another client sends a request, the second client will be placed into a queue since, the server cannot simultaneously send response to more than one client. This might lead to unacceptable delays in response times.

Figure 3.9 shows Request-Response Communication Model.

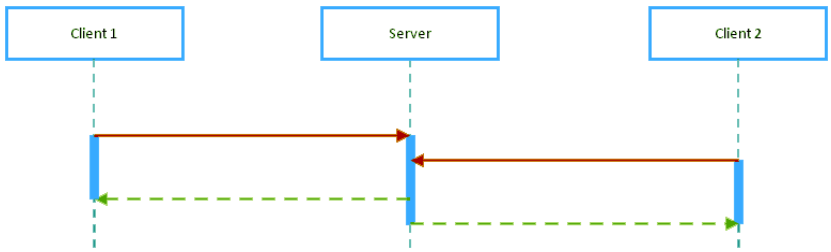


Figure 3.9: Request-Response Model

Subscribe-Notify Model

In Subscribe-Notify model the client specifies the interest in a service on the server by directing a subscribe-call to the server, without having to wait for the server to respond. The server saves the client's subscription together with the address of the client and sends notifications to this address as when the server is ready. The clients will not be placed in queue and can carry on with other tasks and needs to process the notification only when it is received. In this communication model the server can send notifications to multiple clients who have subscribed to similar notifications.

Figure 3.10 shows Subscribe-Notify Communication Model.

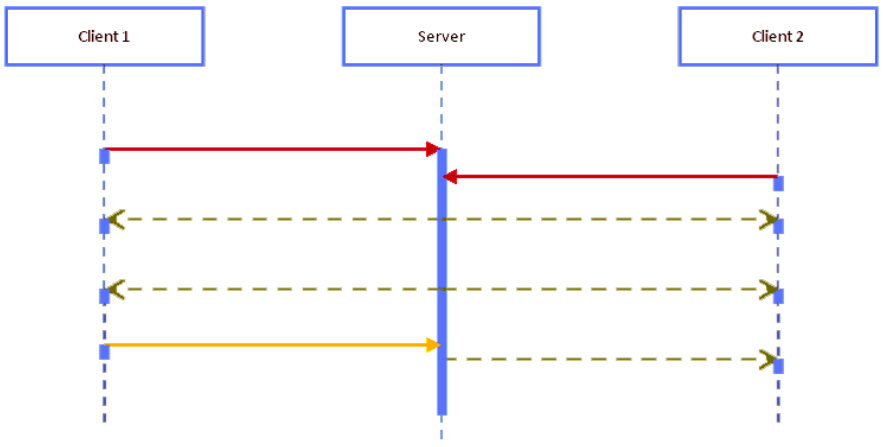


Figure 3.10: Subscribe-Notify Communication Model

Publish-Subscribe Model

In the Publish-Subscribe model there is service providers' information on a broker module. When clients notify their interest in particular information on the broker module, the module will make certain that the information flows between the service and client. Service providers can publish information to the broker irrespective of the number of users interested in this information; if no user has subscribed to the information then the broker does not forward the notification to any user (client). If many clients have subscribed to the same information, the

broker will reproduce the information and send out notification to each subscriber (client). Figure 3.11 shows Publish-Subscribe Models.

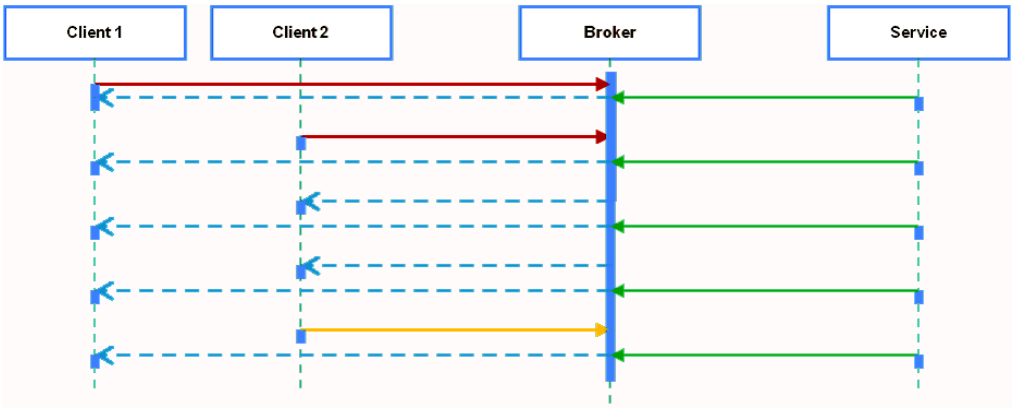


Figure 3.11: Publish-Subscribe Models

3.7 Enabling Devices

IoT is enabled by several technologies including Wireless Sensor Network (WSN), Cloud Computing, Big-data analytics, embedded systems, security protocols, communication protocol, and Semantic Search Engines.

3.7.1 Wireless Sensor Network (WSN)

A Wireless Sensor Network (WSN) is defined 'as wireless network formed by thousands of sensor nodes where each node is equipped with a sensor to monitor physical or environmental conditions such as temperature, humidity, illumination intensity, sound, vibration, pressure, motion or pollutants.' A WSN is equipped with sensing and computing devices, radio transmitter and receiver, battery, electronic circuit interface, and internal and external antenna. WSNs network architectures are of various forms such as star, point-to-point, or mesh network. A WSN comprises number of end-nodes, routers, and coordinator. End-nodes have numerous sensors fixed to them that gather information about the real world and act as routers. Sensor nodes communicate among themselves using radio signals. Routers are responsible for routing the data from the sensors and transmitting it to the coordinator. The coordinator collects data from all the nodes and acts as a gateway or interface that connects users and the network. The advantage of WSNs is easy to deploy, have better flexibility and scalability, and is the key technology for the IoT. WSNs transfer their data through the network to a main location with great reliability and efficiency.

3.7.2 Cloud Computing

Cloud computing is a type of Web-based computing that provides applications and services to users on demand. Cloud computing involves provisioning of shared processing resources and data to users on request. Cloud providers normally use a 'pay as you go' (paying for a service before it is used) model.

Services offered by Cloud computing is as follows:

- **Infrastructure-as–a-Service (IaaS)**: IaaS offers the users virtualized computing resource or computing infrastructure over the Internet such as hardware, software, servers, storage and other infrastructure components.
- **Platform-as-a-Service (PaaS)**: PaaS offers the users applications over the Internet and the ability to develop, run, deploy, and manage these applications.
- **Software-as-a-Service (SaaS)**: SaaS provides software on a subscription basis to users' on-demand and provides access to software functions as a Web-based service.

3.7.3 Big-data Analytics

Big-data refers to data creation, storage, recovery, and examination that is significant in terms of:

- **Volume**: The size of the data determines the value and capacity of the data being considered, after which it can be categorized as Big-data or not. The 'Big-data' term comprise the words 'big and data'.
- **Variety**: This refers to the group to which Big-data belongs. This assists users who analyze the data and are related to it, to efficiently use the data for their benefit and thus, maintaining the significance of the Big-data.
- **Velocity**: This denotes the swiftness of the production and processing of the data to meet the needs of the users.
- **Variability**: This denotes to the discrepancy shown by the data sometimes thus, obstructing the process of controlling and managing the data effectively.
- **Veracity**: The value of the data being captured can differ significantly. Precision of study of the data hinges on the veracity of the source data.
- **Complexity:** Huge volumes of data need to be connected, linked and interrelated so as to enable to store the information. This state is characterized as the complexity of Big-data.

Operations of Big-data

Big-data functions vary for every single IoT device it is connected to capture and store all the incoming data. This information will be examined at single point center to optimize the performance and process. All the processing data is centred on three major steps such as data ingestion, data storage, and analytics. These data is processed with new technologies such as Hadoop, Map Reduce, and so on, which allow the entire device connected. Large volume of the IoT will drastically change the environment of Big-data, as many devices are linked and a lot of information needs to be examined. When this happens, better and faster processing technologies should be introduced to allow and analyze all these information.

Advantages of Big-data are as follows:

- Big-data can give precise access to more data unlike before. Unstructured data would have been considered 'dead' and of no value, whereas with Big-data it can be collected and scrutinized. It gives the opportunity to ascertain data correlations and patterns that would have earlier been hidden persistently, using some application. This allows organizations to access more accurate information.
- Big-data analytics is helping the companies to make investments in implementation of new service models for its industrial merchandizes. The most important use of Big-data analytics is to produce new products and services for customers.
- Big-data can lead to quicker, as well as, enhanced decision. Big-data is not only a method of storing petabytes or exabyte of data, it is also about capacity to make better decisions and take actions at the accurate time.
- Big-data technologies such as Hadoop and cloud-based analytics can provide considerable cost advantages. The drawback with traditional relational database management systems is that it is very costly to scale to such a degree in order to process such enormous volumes of data. However, Hadoop is designed as a scale-out architecture that can reasonably store all of a company's data for future use.

3.7.4 Embedded Systems

An embedded system is a computer system that has computer software and hardware embedded in it, to execute certain functions. Hardware components connect IoT devices to the real world and perform several complex and intricate tasks. The software in the embedded systems interprets data

or information that is transmitted into the embedded systems from other connecting devices and controls the embedded systems. Embedded systems, such as, sensors, microcontrollers, and small chips, when implanted in machines or devices, supervise several functions of the machines. Embedded systems can be found in machines, such as, digital watches, digital cameras, POS, audio/visual systems, DVD players, automobile tools, vending machines, avionics, thermostats, lightings, and so on.

3.7.5 Communication Protocols

The IoT is based on the networking of things. IoT is defined as a 'proposed development of the Internet, in which everyday objects have network connectivity, allowing them to send and receive data.'

The important wireless communication protocols of the IoT are as follows:

- **Satellite**: Satellite communications permit mobile phone to send and receive data through the mobile network. The advantages of this communication protocol are stable connection and global compatibility.
- **WiFi**: WiFi is a Wireless Local Area Network (WLAN) that uses the IEEE 802.11 standard. The advantages of WiFi are smartphone compatibility, affordability, and it is well protected and controlled. WiFi connect to an external cloud-server and are not directly connected to the smartphone.
- **Radio Frequency (RF)**: RF communications are the simplest form and uses low energy consumption to transmit and receive radio signals between two devices. However, a gateway node is needed to link the RF devices to the Internet.
- **RFID**: Radio Frequency Identification (RFID) does not require power and uses radio wave technology to identify objects. RFID is an established and widely used technology; however, it is not compatible with smartphones.
- **Bluetooth**: Bluetooth is a wireless technology for exchanging data between two devices over small distances and which are in proximity to each other. The advantages of Bluetooth are each smartphone has Bluetooth and when the technology in smart phones is constantly upgraded Bluetooth can adapt to such changes in technology.
- **Near-Field Communication (NFC)**: Near-field communication (NFC) short-range high frequency wireless communication technology that enable two electronic devices to exchange data between them. NFC uses wireless RFID technology and enables a device to gather and interpret data from another closely located NFC device. NFC is mainly used in contactless payment systems, where there is physical touch between devices or devices are brought very close to each other.

3.7.6 Semantic Search Engines

Semantic search is an information searching method in which a search inquiry not only identify keywords but also understand the purpose of the search and how the data is related (the context), within the same site and throughout the Web. The purpose of the search clearly states what an individual is looking for. Thus, by connecting purpose and context, search engines responds to different queries and meet users' expectations of information or data needed. Examples of Semantic search engines are Bing, Google, CMANTIK, iGlue, Kosmix, Twinword Finder, and so on.

3.7.7 Security Protocols

Security protocol carries out security-related function by protecting data during network transmission. The cloud interface must be protected, since, poorly secured cloud interfaces are easy to notice. Data traffic between a device and the cloud should be inspected to make sure it is secured

and in case of any problems it can be rectified. To guard against this risk, sensitive information encryption should be used in the update process.
Devices should be frequently updated since, risks come from peripheral sources, such as, USB ports, SD cards or other storage means, which can give way to access the stored data on the device.

3.8 Computing Devices

Computing device is a device that has a hardware and software capable of operating, executing and providing services, and applications. The hardware and software of a computing device comprises processors, random memory and storage, Wi-Fi, and a base Operating System (OS).
Figure 3.12 shows various Computing Devices.



Figure 3.12: Various Computing Devices

3.8.1 Raspberry-Pi

The Raspberry-Pi is a computer that is the size of a credit card or an ATM card that can be plugged into a computer monitor or Television, and uses any standard Universal Serial Bus (USB) computer keyboard and mouse.
Raspberry-Pi is very affordable and enables users to study computing and to write programs in software languages such as Scratch and Python by using the specialized capabilities of the Raspberry-Pi device. The device is a kin to a desktop computer and performs all functions that desktop computer does such as creating spreadsheets - Google documents, MS Excel, Open office and Apple iWork, playing High-Definition (HD) videos and audios, browsing the Internet, and Word-Processing - MS Office, Open Office. Raspberry-Pi has been used in a varied digital maker projects.
Figure 3.13 shows a Raspberry-Pi Model.

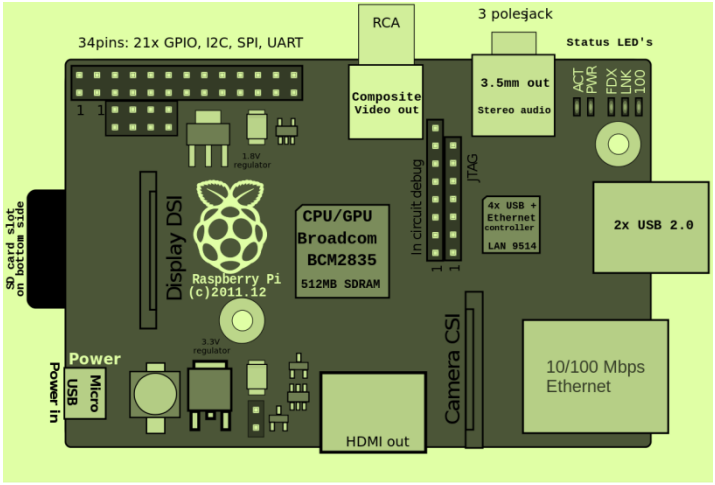


Figure 3.13: Raspberry-Pi Model

3.8.2 BeagleBone Black

The BeagleBone Black is used as an educational board by colleges around the world to educate students on open-source hardware and software capabilities. BeagleBone Black, designed with open source software and is a low-power open-source hardware single-board computer.

Figure 3.14 shows the BeagleBone Black.

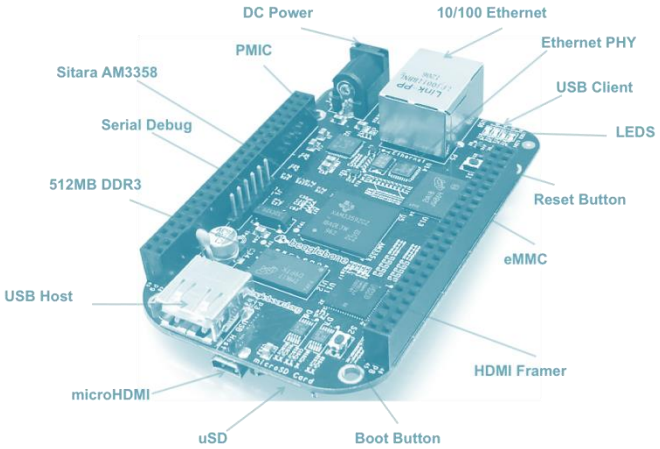


Figure 3.14: BeagleBone Black

3.9 Mobile Phones and Tablets

A mobile phone is a portable or movable communication device that can make and receive calls over a RF transporter while the user is within a mobile phone service area. In addition to making and receiving calls, mobile phones support varied services such as Short Service Text Messaging (SMS), Multi Media Message (MMS), E-mail, Internet access, Wireless communications (Bluetooth), digital photography, games, and applications.

A tablet computer, commonly known as tablet, is a mobile computer with a battery, integrated circuit, and a touchscreen display substituting the use of the keyboard and mouse. Tablets have features such as cameras, microphone, screen pop-up, and virtual keyboards for keying. Tablets also have physical buttons for certain features such as adjusting the speaker volume, power ON/OFF, and ports for network communications and charging the battery. Some tablets also have physical keyboards.

3.10 Wearables

Wearables are small electronic devices integrating computer and advanced technologies that are worn with clothing or on human body. Wearables perform functions of real-world and interweaves with users' everyday life. Wearable devices are IoT devices, embedded with microchips, or sensors, or microcontrollers and enables transmission of data between the 'wearer of the devices' and the 'wearables', without human intrusion or human input. Examples of wearables are smart watches, Google glass, fitness device, wearable headsets, and so on.

Figure 3.15 shows Examples of Wearables.



Figure 3.15: Examples of Wearables

3.11 IoT Platform

The aim of any IoT device is to connect with other IoT devices and applications (cloud-based) to transmit information using Internet TPs. The space between the device sensors and data networks is sealed by an IoT Platform. The IoT platform links the data network to the sensor arrangement and offers insights using backend applications to make sense of the huge volume of data produced by numerous sensors. IoT platform links or is a bridge between the hardware devices and various application layers, thus, enabling intercession between the devices and application layers.

The IoT platforms are capable of being assimilated with any connected device and meld in with the applications used by the device. This independence permits a single IoT platform to execute IoT features into any kind of connected device in the same way.

Figure 3.16 shows a model of IoT platform.

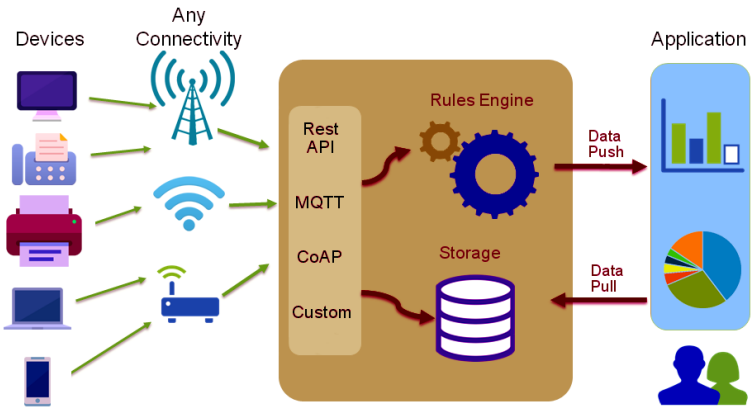


Figure 3.16: A Model of IoT Platform

3.12 Summary

- IoT devices have unique identities and can perform remote sensing, actuating and monitoring functions. IoT devices can exchange data with other connected devices, applications (directly or indirectly), and collect data from other devices.
- Radio Frequency Identification (RFID) uses a physical field produced by electrically charged objects (electromagnetic fields) to automatically or mechanically identify/recognize and track tags affixed to objects. The tags comprise electronically stored data.
- The IoT will need a network that can manage augmented demand for data analytics, dexterity, and safety. Three areas of the IoT that will influence the network are data analytics, the necessity for network dexterity, and safety. To create the right networking technology for applications, it is important to first comprehend the network architecture that is backed by each technology standard.
- A network protocol outlines rules and principles for transferring data/messages between network devices. Protocols for computer networking commonly use packet swapping methods to transmit and accept messages in the form of packets.
- Logical design of IoT denotes the conceptual or intangible representation of the individuals (entities) and processes without going to the low level details of the implementation. An IoT system comprise number of functional blocks that provide the system abilities for communications, identifications, actuating, sensing and management.
- The four types of IoT communication models are - Push-Pull model, Request-Response model, Publish-Subscribe model, and Subscribe-Notify model.
- IoT is enabled by several technologies including Wireless Sensor Network (WSN), Cloud Computing, Big-data analytics, embedded systems, security protocols and architectures, communication protocol, Web services, Mobile Internet, and Semantic Search Engines.
- Computing device is a device that has a hardware and software capable of operating, executing and providing services and applications. The hardware and software of a computing device comprises processors, random memory and storage, Wi-Fi, and a base Operating System (OS).
- The Raspberry-Pi is a computer that is the size of a credit card or an ATM card that can be plugged into a computer monitor or Television, and uses any standard Universal Serial Bus (USB) computer keyboard and mouse. Raspberry-Pi is very affordable and enables users to study computing and to write programs in software languages.
- The BeagleBone Black is used as an educational board by colleges around the world to educate students on open-source hardware and software capabilities and is a low-power open-source hardware single-board computer.
- An IoT platform is defined as an IoT middleware that acts as a mediator between the hardware and application layers. The IoT platforms are capable of being assimilated with any connected device and meld in with the applications used by the device.

3.13 Check Your Progress

1. Radio Frequency Identification (RFID) Tags use _____ to identify tags.
    a. Electricity
    b. Electromagnetic Fields
    c. Cable
    d. Battery

2. Which of the following IoT Network Architecture can be used to cover a broad range or area?
    a. Point-to-Point
    b. Star
    c. Ethernet
    d. Mesh

3. Near Field Communication is a _____.
    a. Security Protocol
    b. Network Protocol
    c. Communication Protocol
    d. Embedded System

4. Beagle Boneblack and Raspberry-Pi are _____.
    a. Computing device for learning Programming Language
    b. Platform for IoT
    c. Communication Model used in IoT
    d. Transmission of IoT

5. IoT application platforms utilizes the power of _____ to take device data points and translate it into meaningful information.
    a. Wireless Sensor Networks
    b. Cloud Connectivity
    c. Ethernet
    d. Web Services

3.13.1 Answers

1	**B**
2	D
3	C
4	A
5	B

SESSION 4 - SECURITY AND PRIVACY CONCERNS IN IOT

Session Objectives:

The session explains issues related to the privacy and security of IoT, the challenges involved, and the future of IoT in the present world. The session further discusses in-depth about subscription based charges.

At the end of this session, students will be able to:

- Describe the role of IoT in the future
- Explain the Security and Privacy issues and Challenges involved
- Explain Subscription based charges
- Explain the concept of Trust in IoT

4.1 Introduction

IoT is an information network connecting virtual and physical objects. Disagreements over the technical and regulatory solutions for the IoT seem to be the main contention in the IoT. These issues will have unfathomable impact on all aspects of users' life. The flaws in the privacy and security concerns and issues of the IoT outweigh the huge potential of IoT in several areas and are susceptible to loss of privacy for the end users.

The applications of the IoT are closely linked to sensitive infrastructures and strategic services, such as, water and electricity, surveillance of assets, and so on. Privacy and the levels of security that IoT guarantees to systems and processes only will bring about confidence and acceptance of IoT. IOT needs to provide protection to its users' due, its uses in sensitive information about users' location and movements, users' health and purchasing preferences, and so on. Some of the important security and privacy issues plaguing the emerging IoT system, comes from the security issues that are in-built in the technologies used in IoT. These IoT technologies are used for information and data communication from one device to another.

Objects and environments provide information to users about their state or their surroundings and can be used remotely. IoT enables people and objects in the physical and virtual environments to interact with each other so as to create smart environments such as smart health systems, smart cities, smart energy systems, and so on. IoT will improve the quality of users' life, create new markets and new jobs, increase economic growth, and increase competition.

IoT privacy and security issues are the special considerations essential to protect the information of users from exposure in the IoT environment, in which any physical or object can be given a unique identifier and the ability to communicate freely over the Internet or any other similar network. The communication infrastructure is based on an extension of the Internet, which enables transparent use of the object resource across the globe.

Smart objects densely populate users' life and environment, interacting with both by providing, processing, and delivering any sort of information. The idea of linking appliances and other objects is

comparatively new, especially in terms of the global connectivity and independent data transfer that are central to IoT. However, security has not been considered in product design, which can make daily life household objects subject to vulnerability. Sensors are integrated in buildings, vehicles, attached to animals, implanted into human.

These sensors communicate among them locally and remotely in order to provide integrated services. The deep penetration of technology that introduces a new kind of automation and remote interaction has raised new security and privacy issues.

The 'things' in the IoT environment transmits data autonomously and works in conjunction with 'other things' and communicates with them. Interoperability (The ability to exchange and make use of information) of things is essential to the IoT's functioning. The data transmitted by a given 'thing' might not cause any privacy issues on its own.

However, when fragmented data from multiple 'things' is collected, organized, and scrutinized, it can produce sensitive information. Michael Jordon, research director at Context Information Security (CIS) commented on the need to tackle the security and privacy issues related to the IoT: 'IoT security needs to be taken seriously, particularly before businesses start to connect mission critical devices and systems'

Figure 4.1 shows the IoT connecting users' through cloud.

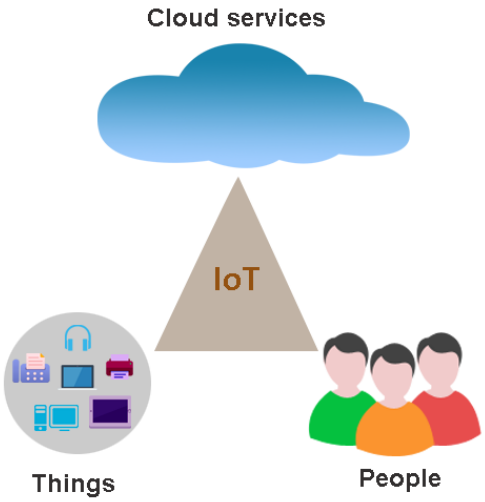


Figure 4.1: IoT Connecting People Through Cloud

4.2 Future of IoT

As technology is becoming progressive and access to the Internet possible at each corner of the world, the prospect of IoT is certainly bright. Achieving greater connectivity is the prerequisite for development in the present world; thus, the IoT. The IoT has become a major channel for interconnecting devices. IoT will transform users' life beyond imagination. It would make life faster, easier, and more productive. Users want to have far-reaching access to all products at their homes when they are outside.

The IoT is changing the daily life physical objects that surround us into a system of information that will develop users' life. The IoT extends communication past the traditional borders. Refrigerators,

washing machines, vending machines, and containers, which were considered as irrelevant to communications, can presently be networked with smart communication components. The number of connected devices being added into communication network is increasing rapidly, which will result in ubiquitous network taking shape and growth.

In the present phase of the IoT, most applications are independent and set up for specific users. The growth and maturity of distributed intelligent information processing technologies, has led to the IoT making intelligent sensing extensively available through information sharing and collaboration. In the near future, real world things such as vehicles, homes, machines, and tools will be connected to the Internet in order to make users' life safer, comfortable, and environmentally friendly.

The IoT will have an overwhelming impact on users' daily life and will play a vital role in the development of key infrastructures, such as, electricity, health care, retail, water resources management, petroleum, transportation, industrial control, and utilities management.

The IoT will greatly improve productivity and users' life. The IoT's huge market potential will attract major investments from telecom, manufacturers, governments, and manufacturing users.

IoT will lead to better understanding about environmental and social issues as increasing users will have access to the Internet and thus, will have access to new techniques and solutions for education, environmental hazards, and health hazards. IoT in the future will bring the following changes in users' life:

- **Monitoring and Reporting**: Millions of devices and equipment will be connected with each other with the IoT concept. Using the IoT, each and every movement in users' home can be monitored and connected to the Internet. The reporting and monitoring of home through Wireless Sensor Networks (WSNs) can help users to take preventive measures, if any accidents or hazards are sensed by the WSNs. Smart cities will ensure clean and safe environments, intelligent smart traffic systems, thus, guaranteeing lesser accidents.
- **Plants and Animals**: IoT will initiate remarkable innovation in the way food is grown, processed, distributed, stored, and consumed. Users will be able to identify the requisites of plants and animals, based on the information that will tell users, computers, and machines as to when; plants and animals need water, treatment, healthcare support, more sun or individual attention.
- **Utility:** IoT will lead to huge technological development in all spheres and fields. It will make daily life simpler with availability of technology, which will be user friendly. For example, Power companies will read meters through smart meter systems instead of visiting houses, vehicle mounted terminals automatically shows the nearest parking space and sensors embedded in smart homes will automatically turn OFF lights, shut windows, monitor and oversee security, and report to home owners in real time; doctors will remotely monitor the health conditions of their patients 24x7 by allowing the patients use devices at home instead of the patients staying at the hospital.
- **Information**: In future, Internet access will be much faster than expected or imagined. Access to Internet anywhere, anytime, without any expensive gadgets will be made possible. Consumer applications will drive the number of connected things, while firms will account for most of the revenue. Gartner, an information technology research based advisory firm, estimated that 2.9 billion connected things are in use in the consumer sector in 2015 and this will increase to over 13 billion till 2020.

Figure 4.2 shows the future of the IoT.

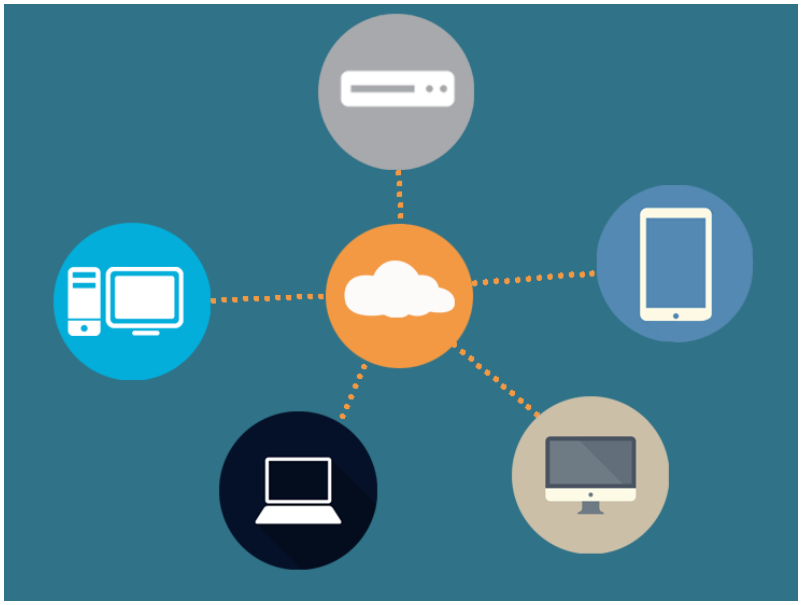


Figure 4.2: Future of IoT

4.3 Subscription Based Services

A subscription-based pricing model is a 'payment structure that allows a customer or organization to purchase or subscribe to a vendor's IT services for a specific period of time for a fixed price. Subscribers typically commit to the services on a monthly or annual basis'.

A lot of hype surrounds the IoT, with many firms trying to distinguish themselves in an apparently crowded space. In the retail industry, many firms advertise neat and transitory products, services, and ideas which they hope will encapsulate the imagination of the users who purchase them before their thoughts change course. There need not be a justified consumer (user) expense, as users are free to spend their money as they wish.

The major business enabler of the IoT is the wealth of information and data that has become accessible. For example: monitoring heavy equipment, checking employee productivity on the assembly floor, constructing a highway through the mountains, or handling a fleet of vehicles, the IoT has made it feasible to monitor operational and diagnostic information in real-time. Some of this information is ordinary and does not need immediate attention, but some of it has the ability to transform businesses.

A new business model has emerged lately that is in the centre of transforming how every business functions the '-as-a-Service' model. The customary hardware business model has been built on selling users a device, with a warranty or guarantee, and then completing the business deal. Firms are now able to use the functional and analytical information from these devices to provide added value over the lifespan of the device, or closely bind the user to the firm. This is facilitated through the 'as-a-Service' model, which is centred on a pay-per-month/use type of business. For example, New Smart Metering is the 'as-a-Service' offering.

This '-as-a-Service' model works quite well for both offering and subscribing firms equally. Instead of budgeting for and spending large capital expenditure for equipment, firms can choose to go with a distinct provider that provides a product with regular, foreseeable operational expenses.

If the service becomes inferior, the purchasing firm has no responsibility, or equity, tied to the provider firm. This type of deal places the onus on the provider firm to maintain a high standard of service.

For the provider firm, this type of deal will decrease the erratic nature of revenue highs and lows by offering means for generating regular revenue. Users today are more knowledgeable and demanding than they were few years ago. They have a huge wealth of information and knowledge at their fingertips. They anticipate on-going value and inimitable experiences. Users are not as interested in procedures as they are in outcomes.

Stand-alone products do not appeal anymore. To meet the anticipations of today's users, firms must go further than products into services from a product experience to a subscription experience. They have to provide services that can study and adapt centred on behavior, services that can develop themselves independently, and services that can genuinely be customized.

4.3.1 Pay-As-You-Go Model

Pay-As-You-Go (PAYG) cloud computing is a payment system for cloud computing that charges based on usage. The system is analogous to that of service bills, utilizing only resources that are required. The primary benefit of the PAYG method is that there are no unused resources, since users only pay for services obtained, instead of setting aside a particular amount of resources that may or may not be utilized.

With outmoded enterprise architecture, users' design data storage to manage the maximum work capacity. The public cloud which is a PAYG system permits users to be charged only for what they store. Users choose the CPU, OS, Networking capacity, memory, security, storage, and access controls, and any supplementary software required to manage their environment.

PAYG is a service computing payment method that is executed in cloud computing and targeted at establishments and end users. A PAYG user is billed for utilized, instead of actual computing resources. PAYG enables a user to scale, modify, and set aside computing resources, comprising software, development platforms, and storage. Resource charges are built on utilized services, versus a whole infrastructure. Public cloud resources manage and use the PAYG model in different ways.

4.4 Privacy and Security Issues and Challenges

IoT envisages as a universal network, connecting any objects around us, varying from home appliances, wearable things, industrial applications, and so on. With IoT infrastructure, physical objects such as wearable objects, TV, refrigerator, smart phones, inventory, lightings, supply chain items, and any objects across the sphere will get connected using the Internet.

Sensors, RFID, mobile technology, embedded or implanted systems and Internet technology and knowledge will play crucial roles in IoT infrastructure. Security and privacy issues in IoT situations will be more challenging than the conventional wireless conditions.

Figure 4.3 shows statistics of IoT Privacy and Security issues.

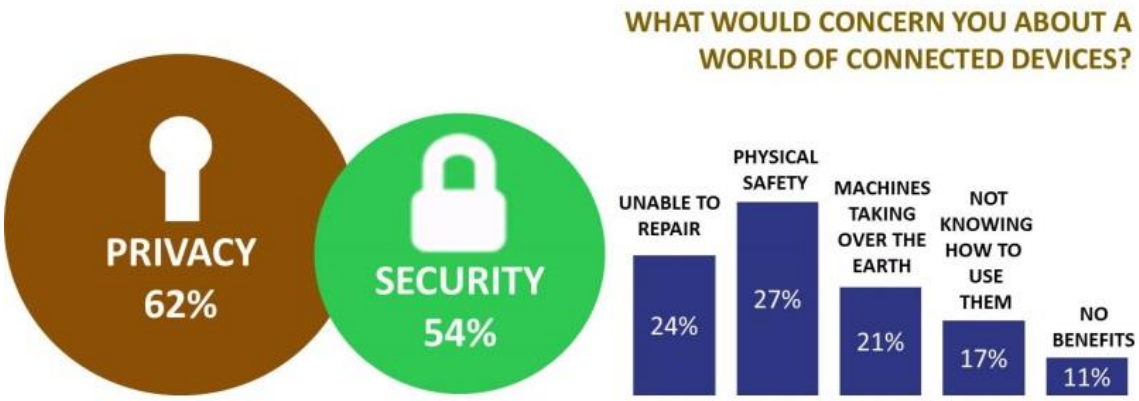


Figure 4.3: Statistics of IoT Privacy and Security Issues

4.4.1 Privacy for IoT

It is important to equally balance advantages of the IoT and users' privacy and security, the development of the IoT is expected to be gravely compromised. The on-going use of IoT devices is currently creating serious issues related to the privacy of users, on the IoT security, and the possible threat of cyber criminals controlling sensors and smart devices connected to the Internet. The industry is keen to applaud the advantages of the IoT and its prospective impact on the economy. However, to maintain this business, governments, and companies need to analyze their privacy and security measures while deploying the IoT.

As devices become smaller, they tend to distribute more quickly. The projected IoT devices within the next few years - a majority of them will be sensors. These sensors and other devices could generate a vast range of information about users' activities. The IoT means things that can sense – things that encapsulate specific data, transfer such data, and contribute to systems that scrutinize such data. The IoT devices gather data in the form of miniature records of specific sensory states. Millions of these tiny, individual data combine to portray an image – an actual unit of information. It is in this information that the real value, and danger, of IoT lies. IoT enables numerous devices to connect to the Internet, network, and inter-operate in order to make users' life more efficient, convenient and cost-effective.

The IoT will expedite the life of the users but with such facility to users', there will be new privacy threats. In the year 2008, the U.S. National Intelligence Council (NIC) cautioned that the IoT will be an interruptive technology by 2025. The chances of public devices getting connected to networks linked to critical systems, can lead to attacks by adversaries, thus, leading to threats to the nation's security. According to this fact business leaders believe that issues related to data privacy are hindering consumer's acceptance of the IoT.

The IoT creates privacy issues that device manufacturers and application developers have not predicted. The devices that are a part of the IoT enable the storing, analyzing, controlling, and distribution of huge quantities of data with other integrated devices and users. Users' privacy is threatened because of their restricted control and choice over the gathering, retaining, and dissemination of their data. The risk of an incompetent legal structure monitoring the IoT needs improvement in legal analysis and new approaches in regulation.

Major privacy threats related to the IoT are as follows:

- **Unlawful Surveillance**: The Internet-connected components fitted in various objects, such as, cars, homes, and home appliances, can be subjected to unauthorized surveillance. The Internet-connected components enable criminals to receive much more information than they would have otherwise got hold of. For example, criminals will be able to watch children through smart cameras fitted in their toys, monitor users' movements through Internet components installed in their wearables, and monitor when a user enters or leaves home by hooking up to an Internet-connected door lock. Such risks are not merely abstract. In several Internet-connected components fitted in cars, wearables, and toys, there are real vulnerabilities that can pose a potential threat to privacy.
- **Active Intrusion in Private Life:** The Internet-connected components will enable criminals not only to inertly monitor their victims, but also to vigorously intrude in users' private life. This is because many of the Internet-connected components are fitted in objects that can be remotely operated. For example, a criminal will be able to remotely start or stop the home appliances, and unlock the door. Security researchers have found security weaknesses enabling criminals to hack various Internet-connected devices. Hacked and remotely operated home appliances will cause deep stress for the users.

  Study has revealed that for a user, victim to hacking on Internet-connected home appliances, the attack will be similar to the 'poltergeist phenomenon'. The 'poltergeist phenomenon' is described as inexplicable sound or appearance of sounds or motion of objects.
- **Data Profiling**: Data profiling is the collecting, accumulating, and organizing of data about users in databases, which can be used to identify, separate, and classify to make decisions about users known to the decision maker through users' computerized profile. The undisclosed information submitted by the Internet-connected components can be used for a creation of in depth profiles of the users of those components. In turn, the profiles can be used for targeted marketing.

  The term 'targeted marketing' refers to marketing in a way so as to reach users based on different behavioral, demographic, and psychographic characteristics. For example, users who store a large quantity of milk-based products in a refrigerator that has been automated (programmed) to identify the products being stored inside will receive more advertising about milk products than other users. The targeted advertising will create a threat to personal independence. The users may not want to receive advertisements related to their preferences. Profiling is based on undisclosed data. Most data protection laws do not apply to undisclosed data. This means that a user whose data is accumulated for profiling has no ways to find out about the profiles made on the basis the data.

Figure 4.4 shows the privacy risks exposed to users.



Figure 4.4: Privacy Risks Exposed to Users

4.4.2 Trust in IoT

The number of connected devices is increasing and the amount of data they generate is increasing even faster. There are remarkable connectivity technologies in the current real world and new ones are developing. The main elements of trust in IoT are as follows:

- **Accessibility**: The need for accessibility varies for different uses of the IoT applications. In some cases the lack of access will not result in serious consequences, but in others it could lead to grave consequences. However, the user needs to have a logical expectation of the accessibility of the network. For example: Mobile network has by far the biggest coverage universally and it has very high accessibility.

  However, the mobile service promise is not all the time available or accessible universally. The main reason is limitation in its architecture that allows only for a single point of malfunction. In a disintegrated mesh network, the single point of breakdown is circumvented by design. Single node malfunction will not result in a system wide breakdown because other nodes will continue to function and distribute data. This is a big benefit in serious use cases.
- **Privacy and integrity**: The amount of connected devices will be enormous in the future. However, the challenge to identify, monitor, validate and authorize data properly will be important. For managing data, the context where the data is generated needs to be considered to conclude whether it creates any privacy issues or not.

  IoT networks transmit complex data that needs high privacy. Privacy in terms of data safety is required when users consume electricity, lightings, or water, and so on. The third element of trust is reliability. In order to be sure if the data is correct and that no one has deployed the data for individual benefit or for causing harm, reliable connectivity and data management is vital.

IoT will create a world where physical objects are effortlessly assimilated into information networks to provide progressive and intelligent services for users.

Trust management plays a significant role in IoT for consistent data fusion and data mining, competent services with context-awareness, and improved user privacy and information security. It helps users to overcome uncertainty and risks and take part in user approval and utilization on IoT services and applications. The advantages users' get from trillions of smart devices is expected to outweigh the troubles of an intermittent failure and the problem of enduring maintenance and upkeep.

If problems with security, performance, or privacy are discovered the issue can be rectified with an s/w patch. If the issues are fixed promptly, defined convincingly, and simple to execute without major interruption, users' will continue to use the devices. However, if the trust breaks in users', then IoT devices will lose its prominence in terms of reliability and dependency, thus, affecting the continuity of using such IoT devices by users.

Figure 4.5 shows Smart Devices Circle of Trust.

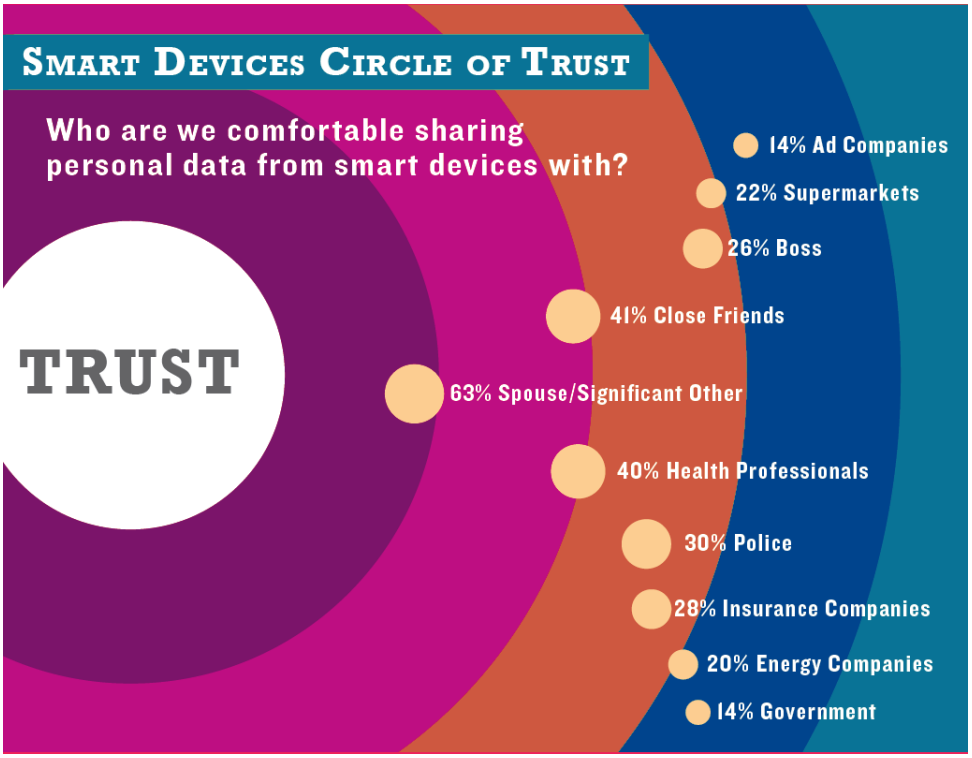


Figure 4.5: Smart Devices Circle of Trust

4.4.3 Security for IoT

IoT security is concerned with protecting connected devices and networks in the IoT. The IoT comprises objects and entities, which are provided with unique IDs and has the capability to automatically transfer data across a network. Some of the major IoT devices are sensors used in industries, smart energy grids, home and building automation, V2V communication, and wearables.

The idea of networking appliances and other objects being relatively new, security aspect has not been given importance in the product design. IoT products are often sold with conventional and unpatched embedded O/S and S/W. Users' often fail to change their default passwords on smart devices or select very strong passwords, if they change.

To enhance security, IoT devices that requires being directly accessible across the Internet, should be divided into segments in its network and have limited network access. The network segment should be controlled to detect potential irregular traffic and remedies should be taken if a problem is detected.

Security experts have cautioned about the possible risks and the increasing number of unsecured devices connecting to the Internet since the conception of the IoT in the late 1990s. The IoT comprising digitally connected devices are impinging on each aspect of users' life, including homes, offices, cars and even users' physical body. The advent of Internet Protocol version 6 (IPv6) and the wide use of Wireless Fidelity (WiFi) networks, the IoT is developing at an alarmingly fast pace, and researchers project that by 2020, the amount of active wireless connected devices will be more than 40 billion. The advantage is that users are able to do things beyond their imagination. The IoT has a major disadvantage; it is becoming an easy target for cybercriminals.

Increase in connected devices means increase in risks of being targeted by hackers or cybercriminals. The imperative need to address this mounting security concern will prevent an inevitable disaster. IoT vulnerabilities will open up new possibilities to hackers and cybercriminals. The increasing

susceptibilities to risk found on IoT devices, will drive IoT security to be upmost in the list of issues to be addressed in the IoT.

Researchers have found critical vulnerabilities in a number of IoT monitors, which can be used to the advantage of hackers, to carry out various unlawful activities, including controlling live feeds, altering camera settings and allowing other users to remotely view and control the monitor.

Similarly, Internet-connected cars can be hacked, as well as, hackers can carry out various malicious activities, including controlling the entertainment system, unlocking or locking the doors even when the car is in motion. Wearables can also be a source of threat to users' privacy, as hackers can use the sensors implanted in smart watches to steal data, such as, health monitoring data, any other data that users have fed into these wearables, and so on. These risks can have a detrimental effect on the users' life and health.

4.4.4 Security Issues in the IoT

A significant number of IoT devices contain varied vulnerabilities, including password security, encryption and lack of smooth user access permissions. Issues and concerns related to mobile security are a challenge in present era of connected devices. The vast growth of IoT devices has offered a potential doorway into IT infrastructure, company, and personal data. These devices range from Webcams, home thermostats, remote power outlets, sprinkler controllers to hubs for controlling multiple devices, garages, scales, door locks, and home alarms. The devices also comprised some form of cloud service and mobile applications that can be used to access these devices remotely. Security concerns for the IoT are developing at a faster pace than the IoT itself. Tackling IoT related concerns requires identifying the issues related to IoT security. The main issues relating to security of IoT are as follows:

- **Privacy Concerns**: A vast majority of the devices gathered some personal information of the users through the device, the cloud or the device's mobile application. The devices gathered information, such as, name, address, and date of birth or health and credit card data. Many of these devices transmit this information over networks without encryption. If users do not configure their home network correctly, then they are at risk of exposing this information through wireless networks. Cloud services, which most of these devices use, are also very vulnerable and easily susceptible to risk.
- **Insufficient Authentication/Authorization**: Many users fail to create passwords of adequate length and complexity. A large number of users and devices depend on vulnerable and easy passwords and authorizations. For example, Many devices and their cloud modules accept passwords, such as, '1234' or '123456'. Users who configured their accounts with vulnerable passwords, use the same password in the cloud for cloud products.
- **Transport Encryption**: Many devices use unencrypted network services. Transport encryption is where information or data that is being transmitted from one device to another device is encrypted from the beginning of the transmission. Transport encryption will be vital since, many devices transmit data or information, which users consider vital. However, most devices inspected failed to encrypt data, while the devices were connected to the Internet. The need for encryption is crucial considering that a huge amount of data or information is being transmitted around the device, the cloud and mobile applications.
- **Web Interface**: The major security concerns are centered on the Web user interfaces. These issues comprise incessant cross-site scripting, inadequate session management and poor default credentials. These issues enabled hackers to identify valid user accounts and misuse them using password reset features.
- **Insecure Software**: A majority of users do not use encryption while downloading software updates. The number of software updates that users download does not match up to the encryption of such software updates being downloaded. Many of these download that were

verified could be interrupted and uploaded into a file system in Linux, where the software could be viewed and modified.

Figure 4.6 shows data of security concerns in the IoT.

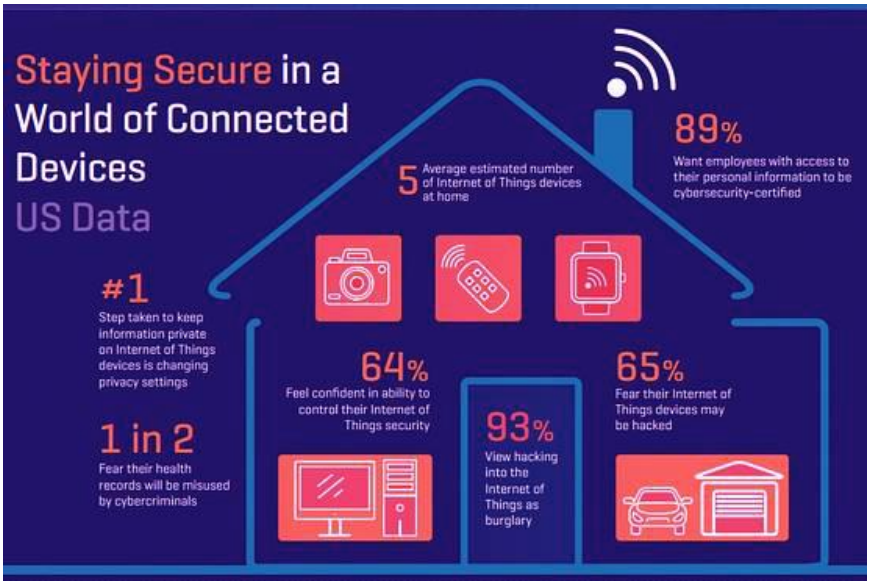


Figure 4.6: Data for Security Issues and Concerns in the IoT

4.4.5 Nobody is Anonymous

There are few firms that discreetly trying to understand the different computing devices that are installed in homes and other places. These firms implant imperceptible sounds into the Webpages that users read and the TV commercials they watch.

Software surreptitiously implanted in users computers, tablets, and smartphones detects and picks up the signals, and then uses cookies to communicate that information or data back to these companies. This helps the firms to track users across different devices. It can connect with things users organize on their tablets and also with things they do on their work computer.

Surveillance is the new business model of the IoT and the more these firms comprehend personal details of users' life, the more they can benefit from it. There are many firms that discreetly spy on users as they browse the Internet, connecting users' behavior on various sites and using that information or data to target advertisements. Users can identify it when they search for something, such as, a vacation, and advertisements for similar vacations follow users around the Internet for days. Companies such as Google and Facebook makes huge profit connecting the things users write about and are interested with firms that are trying to sell users these things.

Cross-device tracking is the latest fascination for Internet marketers. Users utilize multiple Internet devices, such as, computer, smartphone, tablet, Internet-enabled television, and, increasingly, IoT devices, such as, smart thermostats and home appliances. All of these devices are spying on users, but the various spies are mostly unaware of each other. Retailers require this information or data very much.

Retailers want to understand whether their TV advertising make users to look for their products on the Internet. They want to associate users' Web searching on their smartphones with their buying preferences on their computers. They want to track users' locations or places using the surveillance abilities of their smartphones and use that information or data to send geographically or location

wise targeted advertisements to their computers. They want the surveillance data or information from smart appliances linked with each other thing.

The IoT makes the problem worse. Technically, many users give their consent unknowingly by clicking 'I agree' on license agreement, displayed on a screen, or while opening a package they purchase, which they do not read but legally give their consent. This provides all those firms the legal right to conduct surveillance on users. The Internet surveillance economy is less than 20 years old, and developed because there was no directive restricting any of this conduct. It is currently an influential industry, and it is expanding beyond computers and smartphones into each aspect of users' life. The 24x7 surveillance of each thing users do with the implementation of the IoT, is increasing at a fast pace. The development of IoT, results in each thing being monitored, from users' preferences, such as, interest in music, customary habits, carelessness in turning OFF the lights, food and drinking habits, the time spent in chat rooms, Internet downloads, the programs watched on TV and photos taken, the air users' breath, the energy wasted, and so on. There is no upper limit for such things.

4.4.6 Keeping Secrets

The devices in the IoT manage sensitive information or execute vital tasks. Therefore, these devices needs to be trusted, store data, and communicate confidentially. The devices also require proving their identity and verifying messages that they transmit and accept and that have not been tampered interfered with. These vital security abilities all depend on cryptography – and all cryptography depend on secrets.

The use of encryption, digital IDs or signatures is crucial for keeping secrets in the IoT and keeping secrets is a big challenge in the IoT. The IoT developed at a time when risks to data, information and systems have been large.

The advent of new connected devices into the IoT market has resulted in security researchers examining the vulnerabilities of such new devices, in order to make the users aware of the latent risk of connecting the devices, in the absence of proper security. Security of the devices in the IoT is essential in the present interconnected world. IoT devices range from smart TVs, refrigerators, lightings, and home surveillance system to industrial machines, uranium centrifuge control systems, and so on.

The security of connected devices, if compromised, will lead to lack of reliability in using these devices for secure data exchange, processing, or storage. If critical systems and infrastructure, such as, nuclear plants, implantable medical devices, and water and electricity distribution system are hacked, then the universal trust would be affected severely.

4.5 Cryptography

One of the significant methods to preserve secrets in the IoT is cryptography. Cryptography is a method of storing and transmitting data in a particular form so that only those for whom it is intended can read and process it.

Figure 4.7 shows trust in the IoT through Cryptography.

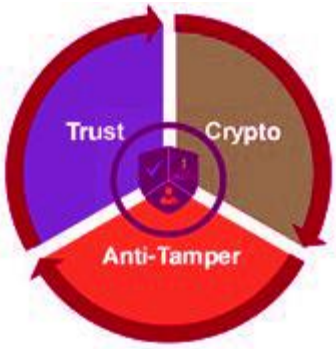


Figure 4.7: Trust in the IoT through Cryptography

Contemporary cryptography concerns itself with the following four objectives which are as follows:

- **Confidentiality**: The information or data cannot be comprehended by anyone for whom it was not intended.
- **Integrity:** The information or data cannot be altered or tampered in storage or transmission from the sender and intended receiver without the alteration being identified.
- **Non-repudiation**: The sender of the information or data cannot deny the intentions in the transmission of the information or data.
- **Authentication:** The sender and receiver can confirm each other's identity and the origin/destination of the information or data.

The requirement for cryptography in IoT comes from two main needs which are as follows:

- **Data Security**: Encryption is an essential requirement to execute any form of authentication and acceptance. When any data or information is transmitted, encryption is mandatory to prevent any third party from accessing and tampering such data or information. Encryption is required to maintain the access to those data or information, even at a local level.
- **Data Privacy**: Data privacy necessitates stronger need for encryption. The system IoT allows a variety of devices to transmit data that is sensitive and confidential. In IoT the amount of data that will be transmitted and delivered is huge. A more careful approach to data privacy will be required and rooted into the IoT; therefore, encryption will be a mandatory requirement.

IoT in general refers to a wide range of devices that can access the Internet. The necessity to access the Internet is linked to several aspects such as need to transmit data, accept command, and export outputs, and so on. There are varying needs and varying grade of privacy and security needed for each device according to its nature. However, the role and few assumptions are common to all IoT devices which are as follows:

- Deal with sensors
- Deal with data
- Security and privacy implications
- Store data
- Transmit data
- Receive data

Encryption is important for sensors because sensors can retrieve information that can give clues to an expert for things that take place outside the realm of the IoT object. Data is the basis to execute encryption. The storing, transmitting, and receiving data needs encryption Security and privacy implication is another reason for which encryption needs to be implemented. The common mistake related to IoT security consideration is that focus is made mostly on a specific aspect of a device.

Viewing a specific aspect of a device is good for implementation, but not good to comprehend security and data privacy concerns. Issue that seems insignificant in a device assume a different role in a context and IoT is all about context.

Cryptography's role, in this context, is to prevent these data to be used for unauthorized and unwanted activities. Cryptography is also the basic tool required to permit data integrity and acceptance. Cryptography is one of the tools to be used to transmit and store data so as to preserve and protect information.

Data Transmission

When transmitting or receiving data, whether if such data is processed outputs or raw data, it is important that the data:

- Comes from a reliable and approved source
- The data has not been tampered with during transmission (Data injections, data forgering, and so on)
- Data is secured from unauthorized access (Data sniffing)
- The data is harmonious with the requests

Encrypting a transmission, permit the data to pass from one point to another point, without third party intervention, thus, avoiding data extrusion. The key provides a basic level of verification and along with this, data encryption can provide defense against infusions of unsolicited data. The disadvantages of encryption are - solidity of the encryption and key exchange. For example, a 40 symmetrical (regular or proportional) encryption key can be easily forced into contemporary computer systems. On the other end, longer encryption key is of no use if the key is found.

Processor Time and Resources

The longer the key, the more time and resources will be taken for encryption. Encryption chipset can solve this issue, although, the issue of key exchange cannot be fully solved with this method. However, use of long keys in encryption is related to political or costs constrain rather than to technical ones. Costs can be reduced by scaling the production, which would make chips inexpensive. An important aspect to be taken care of, is the overload that encryption offers on network package. An encryption protocol adds to some overload to the transmission due to larger packets and the key exchange process will need several exchanges.

The key exchange issue: The other issue is the key exchange. To make encryption, users need to exchange the key with their partner in transmission of data.

The key can be:

- Static
- Dynamic

A static key is simple to implement can be solidified in the solution. Static key is good for storage but not ideal for data communication. However, if the key is discovered the security is at stake.

Dynamic keys are a more secure solution; many protocols depend on dynamic keys for data swap. The problem with dynamic key is related creating the key. A weak protocol can create some knowable keys that can be easily predicted.

Data storage

Data should be protected when transmitting and storing it. Data storage is not easy in the IoT. There are different types of data - permanent, semi-permanent and volatile. Permanent data stored requires a proper storage area. This area can be local or remote (the cloud) in accordance with the data requirements. The more secure solution will be storing data locally in the device. This is a basic approach, since, the safety of the data stored in devices is related to secure access to the device.

If the device is not able to set up a proper verification and approval mechanism to internal resources, data stored locally require to be preserved from extrusion. Encryption is one of the technologies that need to be implemented. Data transfer also has issues relating to the length of the key. Another concern is the capability of the system to clear out physical data transferred from the storage area, in order to avoid high-level data extrusion techniques.

4.6 Digital Signature

A digital signature 'is a mathematical technique used to validate the authenticity and integrity of a message, software or digital document. Digital signatures are based on public key cryptography, also known as asymmetric cryptography'. Using a public key algorithm one can produce two keys that are mathematically connected - one private and one public. To create a digital signature, signing software (email program) creates a one-way hash of the electronic data to be signed. The private key is used to encode the hash. The encrypted hash -- along with other information, such as the hashing algorithm -- is the digital signature.

Public key cryptography verifies the reliability and authenticity of digital content. Reliability means that the digital content has not been altered since it was generated. Authenticity means that the same digital content has been issued by a well-recognized entity. These two fundamental features are offered by the digital signature scheme and are essential so that the digital content can be relied upon by an electronic device.

The integrity or reliability of digital content is assured by a system called the 'message digest', which is a secure hash algorithm. The important and primary property of a secure hash algorithm is that it is difficult to forge digital content that generates a predefined hash value. The outcome is that two different digital contents generate two different hash values.

The chance of having two different digital contents generating the same hash value is nil. Therefore, if some bytes of the digital content are altered or modified, the hash value of the digital content varies or changes. It is not possible to add some bytes to the changed digital content such that, the subsequent hash value will be the same as the original, unaltered digital content's hash value. Therefore, with a hash algorithm protecting the digital content, it is not possible to discreetly alter or modify that digital content.

The authenticity of digital content is assured by the public key-based digital signature or a cryptographic technique. Public-key cryptography is created on pairs of keys - one private key stored secretly and one public key publicly accessible by anyone. The private key is used to sign a digital content. The issuer of the digital content uses its own private key to identify itself as the issuer. The public key can be used by anyone to authenticate a digital content's signature. Those two keys are secured together. Signing content with a private key generates digital signatures that can be effectively authenticated by a public key only.

Digital signature generation comprises two steps. The first step involves hashing the digital content and generating a hash value. The second step involves the hash value, which is 'signed' using the

distinctively owned, anonymous private key of the digital content creator. This second step creates a value (the 'signature') that is fixed to the original digital content.

To authenticate the digital content signature, the first step is, the digital content is hashed again and in the second step, the resulting recreated hash value is used as an input to the signature authentication algorithm, along with the signature fixed to the digital content and the public key. If the algorithm establishes that the signature is genuine, it means that the digital content is matching the original digital content and that the creator of this digital content is indeed who he claims to be.

Public key-based digital signature methods work because the private key is used for signing content by the owner and none other. Thus, the private key has to be kept in safe hands. The public key need not be kept secretly because anyone can validate a digital content's signature. The only essential requirement for a public key is reliability.

The public key is freely available because it gives no sign of the private key; one cannot determine the private key based on the public key. The public key does not permit anyone to perform individually certain actions like signing digital content. However, as anyone can produce a pair of keys there must be a system to authenticate the identity of the public key owner. Therefore, public key reliability, authenticity, and identity must all be ensured.

Figure 4.8 shows the Digital Signature process.

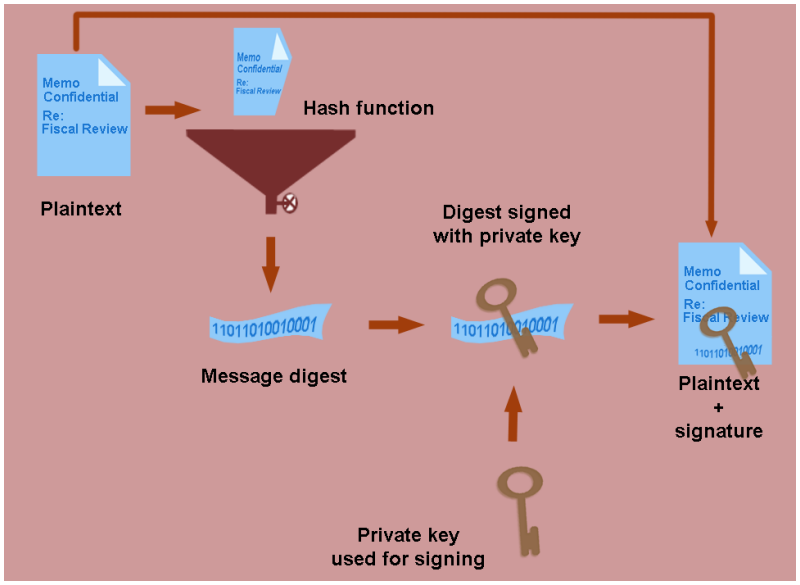


Figure 4.8: Digital Signature Process

4.7 Conclusion

The IoT presents various benefits to users and has the ability to change the ways that users interact with technology in essential ways. In the future, the IoT is likely to fuse the virtual and physical worlds together in ways that are presently difficult to understand. From a security and privacy perspective, the foreseen ubiquitous introduction of sensors and devices, into the personal spaces of users', such as, their homes, vehicles, wearables and the physical body pose specific challenges.

As physical objects in users' daily life are becoming more sensible and share observations and information about them, users require privacy. At the same time, additional self-regulatory efforts on the IoT, along with enactment and implementation of data security and sanctioning of broad-based privacy regulation is needed.

4.8 Summary

- IoT privacy and security issues are the special considerations essential to protect the information of users from exposure in the IoT environment, in which any physical or object can be given a unique identifier and the ability to communicate freely over the Internet or any other similar network. The 'things' in the IoT environment transmits data autonomously and works in conjunction with 'other things' and communicates with them.
- IoT will lead to increased awareness about environmental and social issues, as increasing users will have access to the Internet and thus, will have access to new techniques and solutions for education, environmental hazards, and health hazards.
- The On-going use of IoT devices is currently creating serious issues related to the privacy of users, on the IoT security, and the possible threat of cyber criminals controlling sensors and smart devices connected to the Internet.
- Trust management plays a significant role in IoT for consistent data fusion and data mining, competent services with context-awareness, and improved user privacy and information security. It helps users to overcome uncertainty and risk and take part in user approval and utilization on IoT services and applications.
- Security concerns for the IoT are developing at a faster pace than the IoT itself. Tackling IoT related concerns requires identifying the issues related to IoT security. The main issues relating to security of IoT are privacy concerns, insufficient authorization; Web interfaces risks, transport encryption, and insecure software.
- Cryptography is a method of storing and transmitting data in a particular form so that only those for whom it is intended can read and process it.
- A digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software or digital document. Digital signatures are based on public key cryptography, also known as asymmetric cryptography.

4.9 Check Your Progress

1. Users' privacy is threatened because of their _____ and choice over the gathering, retaining, and dissemination of their data.
    a. Restricted Control
    b. Shortage of connected devices
    c. Unlimited Usage
    d. Free Access

2. Subscription based service in the IoT means _____.
    a. Pay-in Advance
    b. Pay-as-u-Go
    c. Pay-per-Month
    d. Pay-per-week

3. _____ is a key security concern.
    a. Transport Layer
    b. Cryptography
    c. Insecure Software
    d. Digital IDs

4. The requirement for cryptography in IoT comes from _____ and _____.
    a. Data Input and Data Output
    b. Data Transmission and Data Storage
    c. Data Security and Data Privacy
    d. Data Analysis and Data Security

5. Public key cryptography verifies the _____ and_____ of digital content.
    a. Reliability and Authenticity
    b. Technology and Connectivity
    c. Speed and Accuracy
    d. Flow and Transmission

4.9.1 Answers

1	**A**
2	B
3	C
4	C
5	D